



2018. Fall.

1. Let $A \in M_{n \times n}(\mathbb{R})$ such that $A^2 = 0$. Prove

$$2 \cdot \text{rank } A \leq n.$$

Proof. Given $x \in \mathbb{R}^n$, $A(Ax) = A^2x = 0$, so $\text{Im } A \subseteq \ker A$. So, $\dim \text{Im } A \leq \dim \ker A$.
By the dimension theorem,

$$n = \text{nullity } A + \text{rank } A \geq 2 \text{rank } A \quad \square$$

2. Determine over $\mathbb{R}$,

$$A = \begin{pmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix}$$

is diagonalizable.

Proof. The characteristic polynomial of A is

$$\det \begin{pmatrix} t & 0 & 2 \\ -1 & t-2 & -1 \\ -1 & 0 & t-3 \end{pmatrix} = t(t-2)(t-3) + 2 \cdot (-1)(t-2) = (t-2)(t^2 - 3t + 2) = (t-1)(t-2)(t-2).$$

By the Cayley-Hamilton theorem, the minimal polynomial has same roots to characteristic polynomial, so we need to check that $(A-I)(A-2I) = 0$,

$$\begin{pmatrix} -1 & 0 & -2 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} -2 & 0 & -2 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = 0$$

So, the minimal polynomial of A is $(t-1)(t-2)$, it implies A is diagonalizable, because the minimal polynomial splits over $\mathbb{R}$ and separable. Explicitly,

$$\begin{pmatrix} -1 & 0 & -2 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \Leftrightarrow \begin{cases} x = -2z \\ x+y+z=0 \end{cases} \Rightarrow \ker(A-I) = \text{span} \left\{ \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} \right\}$$
$$\begin{pmatrix} -2 & 0 & -2 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \Leftrightarrow \begin{cases} x+z=0 \end{cases} \Rightarrow \ker(A-2I) = \text{span} \left\{ \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \right\}$$

Therefore, $A = UDU^{-1}$ where $U = \begin{pmatrix} -2 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & -1 \end{pmatrix}$ and $D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$.

2018, Fall,

3. Let R be a commutative ring and let $x \in R$ such that $x^{2018} = 0$.
prove that $1+x$ is unit.

Proof. observe that

$$\begin{aligned} & (1+x) \cdot (-x^{2017} + x^{2016} - x^{2015} + \dots - x + 1) \\ &= 1 - x + x^2 + \dots + x^{2016} - x^{2017} \\ & \quad + x - x^2 + \dots - x^{2016} + x^{2017} - x^{2018} \\ &= 1, \quad \text{since } x^{2018} = 0. \quad \text{So, } 1-x+\dots+x^{2016}-x^{2017} \text{ is inverse of } 1+x. \end{aligned}$$

4. prove that A_4 does not have a subgroup of order 6.

Proof. since $|S_4:A_4| = 2$, $|A_4| = 4 \cdot 3 = 12$.

Suppose that A_4 has a subgroup H of order 6.

Then, $|A_4:H| = 12/6 = 2$, so $gH = Hg$ for any $g \in A_4$, it means $H \trianglelefteq A_4$ is normal.
Consider the afforded Homomorphism of A_4 acts A_4/H as left multiplication:

$$\mathcal{Q}: A_4 \rightarrow S_{A_4/H}: g \mapsto (x) \mapsto gxH$$

And, since every 3-cycle $\sigma \in A_4$ satisfies

$$(\mathcal{Q}(\sigma))^3 = \mathcal{Q}(\sigma^3) = \mathcal{Q}(\text{id}) = \text{id}$$

and $S_{A_4/H} \cong \mathbb{Z}/2\mathbb{Z}$ so $\mathcal{Q}(\sigma) = \text{id}$. It implies all 3-cycles are contained in $\ker \mathcal{Q}$. Meanwhile, observe that $(1\ 2\ 3), (1\ 2\ 4) \in A_4$ satisfies

$$(1\ 2\ 3)^{-1}(1\ 2\ 4)(1\ 2\ 3) = (1\ 3\ 2)(1\ 2\ 4)(1\ 2\ 3) = (1\ 4\ 3)$$

$(1\ 2\ 3)(1\ 4\ 3) = (1\ 4)(2\ 3)$, ... in this way, A_4 is generated by all 3-cycles in A_4 , so $\ker \mathcal{Q} = A_4$. By the first isomorphism theorem, $A_4/\ker \mathcal{Q} \cong \text{Im } \mathcal{Q}$ is trivial, so the action is trivial.

But, the left coset action is transitive, it cannot be trivial, contradiction. $\square$

Similar Problems,

1. Show that A_5 has no subgroup of order 15.

Proof. $|A_5| = |S_5|/2 = 60 = 2^2 \cdot 3 \cdot 5$.

Suppose that there is a subgroup $H < A_5$ of order 15.

By the Sylow Theorem, $\# \text{Syl}_3(H) \in \{1\}$ and $\# \text{Syl}_5(H) \in \{1\}$, that is,

there are order 3 subgroup $P < H$ and order 5 subgroup $Q < H$,

and both are normal subgroup of H . Since $P \cong C_3$ and $Q \cong C_5$ both are

cyclic, let $x \in P$ and $y \in Q$ such that $P = \langle x \rangle$ and $Q = \langle y \rangle$.

Consider the afforded Homomorphism by H acts P by conjugation

$$\varphi: H \rightarrow \text{Sym}(P) : h \mapsto (k \mapsto hkh^{-1})$$

Then, $\text{Ker } \varphi = \{h \in H : hkh^{-1} = k \text{ for all } k \in P\} = C_H(P)$ and

$H/\text{Ker } \varphi \cong \text{Im } \varphi \leq \text{Aut}(P)$. But, since $\text{Aut}(P) \cong \mathbb{Z}/2\mathbb{Z}$,

and $|H|$ has no factor 2, odd number, so $|\text{Ker } \varphi| = |H|$, that is,

$C_H(P) = H$. Now, x and y commute, therefore $|xy| = 15$.

But, A_5 cannot have order 15 element, because:

order of permutation in S_5 is determined by the least common multiplier of disjoint cycle decompositions, but $\text{lcm } 5 = 5$,

$\text{lcm}(4, 1) = 4$, $\text{lcm}(3, 2) = 6$, $\text{lcm}(2, 1, 1, 1) = 2$, ...

the lcm cannot be 15, so H contradicts.

[Similar problem

Let $G = S_3 \times C_2$.

1. Find all normal subgroups of G .

2. Prove that G has no normal subgroup of order 4.

Proof. Define three maps:

$$\mathcal{Q}_1: G \rightarrow S_3: (\sigma, \alpha) \mapsto \sigma$$

$$\mathcal{Q}_2: G \rightarrow C_2: (\sigma, \alpha) \mapsto \alpha$$

$$\mathcal{Q}_3: G \rightarrow \{\pm 1\}: (\sigma, \alpha) \mapsto \text{sgn } \sigma \cdot \pi(\alpha) \quad \text{where } C_2 \stackrel{\pi}{\cong} \{\pm 1\}.$$

$$\mathcal{Q}_4: G \rightarrow \{\pm 1\} \times C_2: (\sigma, \alpha) \mapsto (\text{sgn } \sigma, \alpha).$$

$$\mathcal{Q}_5: G \rightarrow \{\pm 1\}: (\sigma, \alpha) \mapsto \text{sgn } \sigma$$

Then, $\mathcal{Q}_1$ and $\mathcal{Q}_2$ are Homomorphisms, being the Coordinatively Projection, and $\mathcal{Q}_3$ is also Homomorphism, because sgn , π are Homomorphism and $\{\pm 1\}$ is abelian. Precisely,

$$\begin{aligned} \mathcal{Q}_3((\sigma_1, \alpha_1) \cdot (\sigma_2, \alpha_2)) &= \mathcal{Q}_3(\sigma_1 \sigma_2, \alpha_1 \alpha_2) = \text{sgn } \sigma_1 \text{sgn } \sigma_2 \pi(\alpha_1) \pi(\alpha_2) \\ &= \text{sgn } \sigma_1 \pi(\alpha_1) \text{sgn } \sigma_2 \pi(\alpha_2) = \mathcal{Q}_3(\sigma_1, \alpha_1) \cdot \mathcal{Q}_3(\sigma_2, \alpha_2). \end{aligned}$$

And, $\mathcal{Q}_4$ is Homomorphism, being entrywise Homomorphism, and $\mathcal{Q}_5$ is Homomorphism, being Composition of Projection and sgn .

Now, the kernel

$$\ker \mathcal{Q}_1 = \{\text{id}\} \times C_2, \quad \ker \mathcal{Q}_2 = S_3 \times \{x^0\}, \quad \ker \mathcal{Q}_3 = A_3 \times \{x^0\} \cup (S_3 \setminus A_3) \times \{x^1\}$$

$$\ker \mathcal{Q}_4 = A_3 \times \{x^0\}, \quad \ker \mathcal{Q}_5 = A_3 \times C_2.$$

are we desired.

Are these all possibilities?

Observe that: if $N \subseteq G_1 \times G_2$ is normal, then the image of projections $\pi_i: G_1 \times G_2 \rightarrow G_i: (g_1, g_2) \mapsto g_i$, $\pi_i[N]$ is also normal of $\pi_i[G_1 \times G_2] = G_i$, because image of normal subgroup is normal in the Range, and the projections are onto.

But, as we know, normal subgroup of S_3 is only $\{\text{id}\}$, A_3 , S_3

and normal subgroup of $\{\pm 1\}$ is $\{1\}$ and $\{\pm 1\}$.

2018. Fall.

7. Consider the identity $\frac{e^{-x}}{1-x} = \sum_{n=0}^{\infty} C_n x^n$.

a). Determine the sequence $(C_n) \subseteq \mathbb{R}$ and the range of $x \in \mathbb{R}$ for the identity holds.

b). Show that for each $n \geq 0$,

$$\sum_{k=0}^n \frac{C_k}{(n-k)!} = 1.$$

Proof. a),

Notice that $e^{-x} = \sum_{n=0}^{\infty} \frac{(-x)^n}{n!}$ for all $x \in \mathbb{R}$ and

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \text{ for all } |x| < 1.$$

Therefore, the coefficient C_n is determined by the Cauchy product

$$C_n = \sum_{k=0}^n \frac{(-1)^{n-k}}{(n-k)!} \cdot 1^k = \sum_{k=0}^n \frac{(-1)^{n-k}}{(n-k)!}.$$

And, the identity valid in $\mathbb{R} \cap (-1, 1) = (-1, 1)$. $\perp 5m$.

b). Since

$$\frac{1}{1-x} = e^{-x} \cdot \sum_{n=0}^{\infty} C_n x^n, \text{ so we can write for } x \in (-1, 1)$$

$$\sum_{n=0}^{\infty} x^n = \sum_{n=0}^{\infty} \sum_{k=0}^n \frac{C_k}{(n-k)!} x^n.$$

And, the coefficients equals for each $n \geq 0$, because these are infinitely differentiable, so the coefficients of power series are determined uniquely by the Taylor Theorem. Thus,

$$1 = \sum_{k=0}^n \frac{C_k}{(n-k)!} \text{ for all } n \geq 0. \perp 4m$$

Tzola, Spring

6. For each $n \in \mathbb{N}$, let

$$f_n(x) = \frac{|x|^n}{1+|x|^n}, \quad x \in \mathbb{R}.$$

a). Show that $(f_n)_{n \geq 1}$ does not converge uniformly.

b) For each fixed $q > 1$, Show that (f_n) converges uniformly on $\{x \in \mathbb{R} \mid |x| \geq q\}$ and $\{x \in \mathbb{R} \mid |x| \leq \frac{1}{q}\}$.

a). Given $x \in \mathbb{R}$,

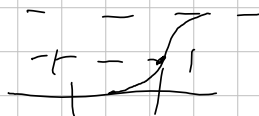
if $|x| < 1$, then $f_n(x) \rightarrow 0$ as $n \rightarrow \infty$, since $|x|^n \rightarrow 0$.

if $|x| = 1$, then $f_n(x) = \frac{1}{2}$ for all $n \in \mathbb{N}$, since $|x|^n = 1$.

if $|x| > 1$, then $f_n(x) = \frac{1}{1/|x|^n + 1} \rightarrow 1$ as $n \rightarrow \infty$, since $1/|x|^n \rightarrow 0$ as $n \rightarrow \infty$.

Therefore, $f_n \rightarrow f$ pointwise where

$$f(x) = \begin{cases} 0 & |x| < 1 \\ \frac{1}{2} & |x| = 1 \\ 1 & |x| > 1 \end{cases}$$



But, for $\epsilon = 1/4$, and for all $n \in \mathbb{N}$,

$$\sup_{0 \leq x < 1} |f_n(x) - f(x)| = \sup_{0 \leq x < 1} |f_n(x)| = \sup_{0 \leq x < 1} \frac{|x|^n}{1+|x|^n} \geq \frac{1}{4}$$

because given $n \in \mathbb{N}$, we can take $x \in [0, 1)$ such that $\frac{x^n}{1+x^n} \geq \frac{1}{4}$

Since $\lim_{x \rightarrow 1^-} \frac{|x|^n}{1+|x|^n} = \frac{1}{2}$, so, $\sup_{x \in \mathbb{R}} |f_n - f| \geq \sup_{0 \leq x < 1} |f_n - f|$ implies

$f_n \rightarrow f$ but not uniformly on $\mathbb{R}$.

b). If $|x| \geq q > 1$, then $f(x) = 1$. So, enough to show that $\sup |f_n(x) - f(x)| \rightarrow 0$.

Since $\sup_{|x| \geq q} |f_n(x) - f(x)| = \sup_{|x| \geq q} \left| \frac{1}{1+|x|^n} \right| \leq \frac{1}{1+q^n} \rightarrow 0$ as $n \rightarrow \infty$, so $f_n \rightarrow f$ uniformly on $|x| \geq q$. Similarly, for $|x| \leq \frac{1}{q}$, $f(x) = 0$ and so

$$\sup_{|x| \leq \frac{1}{q}} |f_n(x) - f(x)| = \sup_{|x| \leq \frac{1}{q}} \left| \frac{|x|^n}{1+|x|^n} \right| \leq \frac{1}{q^n + 1} \rightarrow 0 \text{ as } n \rightarrow \infty,$$

so proved all cases. $\square$

2019, Spring.

17, Let $S = \{(x, y, z) \in \mathbb{R}^3 \mid z = x \cdot f(\frac{y}{x}), x \neq 0\}$ where $f: \mathbb{R} \rightarrow \mathbb{R}$ is differentiable. Show that the tangent plane at any point of S passes through the origin. Prove by the definition of S , given $(x_0, y_0, z_0) \in S$, a map

$X: \mathcal{O} \rightarrow S: (u, v) \mapsto (u, v, u \cdot f(\frac{v}{u}))$ where $\mathcal{O}$ is an open set containing (x_0, y_0) is a parametrization of S at (x_0, y_0, z_0) . The differential of X at (x_0, y_0) is:

$$dX = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ f(\frac{v}{u}) - \frac{v}{u} \cdot f'(\frac{v}{u}) & f'(\frac{v}{u}) \end{bmatrix} \bigg|_{(u,v)=(x_0,y_0)}$$

So, the tangent plane of S at (x_0, y_0) is generated by

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ f(\frac{y_0}{x_0}) - \frac{y_0}{x_0} \cdot f'(\frac{y_0}{x_0}) \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ f'(\frac{y_0}{x_0}) \end{pmatrix} \right\}.$$

Now, for $t_1, t_2 \in \mathbb{R}$, the coordinate of a point of the tangent plane can be described by

$$(x_0, y_0, x_0 \cdot f(\frac{y_0}{x_0})) + (t_1, 0, t_1 \cdot f(\frac{y_0}{x_0}) - t_1 \cdot \frac{y_0}{x_0} f'(\frac{y_0}{x_0})) \\ + (0, t_2, t_2 \cdot f'(\frac{y_0}{x_0}))$$

is zero when $t_1 = -x_0$ and $t_2 = -y_0$, so proved $\square$

201a, Fall.

8. Let $(b_n)_{n \in \mathbb{N}}$ be a decreasing sequence of positive real numbers with $b_n \rightarrow 0$. Let $(a_n)_{n \in \mathbb{N}}$ be a sequence of complex numbers such that $(\sum_{n=1}^N a_n)_{N \in \mathbb{N}}$ is a bounded sequence.

a). Show that $\sum_{n=1}^{\infty} a_n b_n$ converges.

b). Show $\sum_{n=2}^{\infty} \frac{\sin(n\pi x)}{\log n}$ converges for all $x \in [-\pi, \pi]$.

Proof. Denote $A_n = \sum_{k=1}^n a_k$ and $A_0 = 0$. And let $M \geq 0$ be a bound for $\sum a_n$.

$$\begin{aligned} \sum_{n=1}^N a_n b_n &= \sum_{n=1}^N [A_n - A_{n+1}] \cdot b_n = \sum_{n=1}^N A_n b_n - \sum_{n=1}^N A_{n+1} b_n \left(\sum_{n=1}^N A_n b_n = \sum_{n=0}^{N-1} A_n b_{n+1} \right) \\ &= \sum_{n=1}^N A_n b_n - \left(\sum_{n=1}^N A_n b_{n+1} + A_0 b_1 - A_N b_{N+1} \right) \\ &= \sum_{n=1}^N A_n (b_n - b_{n+1}) + A_N b_{N+1}. \end{aligned}$$

$$\sum_{k=n}^m a_k b_k = \sum_{k=n}^m A_k (b_k - b_{k+1}) + A_m b_{m+1} - A_n b_n$$

$$\begin{aligned} \left| \sum_{k=n}^m a_k b_k \right| &\leq \sum_{k=n}^m |A_k| \cdot |b_k - b_{k+1}| + |A_m| |b_{m+1}| + |A_n| |b_n| \\ &\leq M \cdot \left(\sum_{k=n}^m |b_k - b_{k+1}| + |b_{m+1}| + |b_n| \right) \\ &= M \cdot \left(\sum_{k=n}^m (b_k - b_{k+1}) + b_{m+1} + b_n \right) \end{aligned}$$

$$= M \cdot 2 b_n \rightarrow 0 \text{ as } n \rightarrow \infty, \text{ so proved.}$$

b). Since $1/\log n$ is decreasing and positive real numbers, so by a), enough to show that $|\sum \sin(n\pi x)|$ is bounded for all $x \in [-\pi, \pi]$.

observe that: $2 \sin x \cdot \sin nx = \cos(nx - x) - \cos(nx + x)$

$$\text{Therefore, } 2 \sin x \cdot \sum_{n=1}^N \sin(n\pi x) = \sum_{n=1}^N [\cos((n-1)x) - \cos((n+1)x)] \leq |\cos 0| + |\cos x| + |\cos Nx| + |\cos(N+1)x| \leq 4$$

so for $x \in [-\pi, \pi] \setminus \{0, \pm\pi\}$, $\left| \sum_{n=2}^N \sin n\pi x \right| \leq \frac{2}{|\sin x|}$ for any $N \in \mathbb{N}$, bounded.
for $x = 0$ or $\pm\pi$, then the sum is just zero. $\square$

2019, Fall,

1. Let $\varphi: [0, 1] \rightarrow \mathbb{R}$ be a continuous map. Evaluate

$$\lim_{\varepsilon \rightarrow 0} \int_0^1 \left[\frac{\varphi(x)}{x - i\varepsilon} - \frac{\varphi(x)}{x + i\varepsilon} \right] dx.$$

Proof. Observe that

$$\frac{1}{x - i\varepsilon} - \frac{1}{x + i\varepsilon} = \frac{2i\varepsilon}{x^2 + \varepsilon^2}$$

2020. Spring.

6. Find the general solution of the differential equation

$$y'' - 3y' + 2y = \frac{1}{1+e^{-t}}.$$

Proof. Since the characteristic of $y'' - 3y' + 2y = 0$ is $r^2 - 3r + 2 = (r-1)(r-2)$, therefore the general solution of the homogeneous part is:

$$y_h(t) = C_1 e^t + C_2 e^{2t} \quad (C_1, C_2 \in \mathbb{R}).$$

And, to find a solution of the given equation, we need to find the particular solution $y_p(t)$. First, calculate the Wronskian,

$$\begin{vmatrix} e^t & e^{2t} \\ e^t & 2e^{2t} \end{vmatrix} = 2e^{3t} - e^{3t} = e^{3t}.$$

Therefore, the particular solution is

$$\begin{aligned} y_p(t) &= \int_{x_0}^x \begin{vmatrix} e^t & e^{2t} \\ e^{2t} & e^{2t} \end{vmatrix} \cdot \frac{1}{1+e^{-t}} \cdot \frac{1}{e^{3t}} dt \\ &= \int_{x_0}^x \frac{e^{t+2t} - e^{2t+t}}{e^{3t} + e^{2t}} dt \\ &= e^{2x} \int_{x_0}^x \frac{1}{e^{2t} + e^t} dt - e^x \int_{x_0}^x \frac{1}{e^t + 1} dt \end{aligned}$$

Meanwhile, since $\int \frac{1}{e^t + 1} dt = \int \frac{e^{-t}}{1 + e^{-t}} dt = -\log(1 + e^{-x})$

and $\int \frac{1}{e^{2t} + e^t} dt = \int \frac{e^{-t}}{e^{2t} + 1} dt = \int \frac{-u}{\frac{1}{u} + 1} \cdot \frac{1}{u} du = \int \frac{-u}{1+u} du$

$$= \int -1 + \frac{1}{1+u} du = -u + \log(1+u) = -e^{-x} + \log(1+e^{-x})$$

Finally, the general solution is:

$$\begin{aligned} y_p + C_1 e_1 + C_2 e_2 &= \left[e^x + e^{2x} \log(1+e^{-x}) + e^x \cdot \log(1+e^x) \right] \\ &+ C_1 e^x + C_2 e^{2x}, \quad C_1, C_2, C_3 \in \mathbb{R}, \end{aligned}$$

12/20, Fall.

1. Let p be a prime. Find the number of subgroups H of $\mathbb{Z}_{p^2} \times \mathbb{Z}_{p^3}$, $|H| = p^2$.
Proof. By the fundamental theorem of finitely generated abelian group, either $|H| \cong \mathbb{Z}_{p^2}$ or $\mathbb{Z}_p \times \mathbb{Z}_p$.

If $|H| \cong \mathbb{Z}_p \times \mathbb{Z}_p$, then for all $x \in H$, $p \cdot x = 0$. Therefore,
 $H \subseteq \{(a, b) \in \mathbb{Z}_{p^2} \times \mathbb{Z}_{p^3} \mid p \cdot (a, b) = 0\}$.

Meanwhile, $pa = 0$ and $pb = 0$ if and only if $a \in \langle p \rangle \subseteq \mathbb{Z}_{p^2}$ and $b \in \langle p \rangle \subseteq \mathbb{Z}_{p^3}$,
so $\{(a, b) \in \mathbb{Z}_{p^2} \times \mathbb{Z}_{p^3} \mid p \cdot (a, b) = 0\} = \langle p \rangle \times \langle p^2 \rangle$.

Since $\langle p \rangle \times \langle p^2 \rangle$ has already order p^2 , so $H = \langle p \rangle \times \langle p^2 \rangle$.

If $|H| \cong \mathbb{Z}_{p^2}$, that is, H is generated by order p^2 element,
Since $x \in \mathbb{Z}_{p^2} \times \mathbb{Z}_{p^3}$ has order $p^2 \iff p^2 \cdot x = 0$ and $px \neq 0$.

Every element $a \in \mathbb{Z}_{p^2}$ satisfies $p^2 \cdot a = 0$, and $b \in \mathbb{Z}_{p^3}$ satisfies $p^2 \cdot b = 0$
if and only if $b \in p\mathbb{Z}_{p^3}$ and $\mathbb{Z}_{p^3}$ has number of p^2 of p -multiples,
so there are $p^2 \cdot p^2 = p^4$ elements satisfying $p^2 \cdot x = 0$.

And, $px = 0$ if and only if $a \in p\mathbb{Z}_{p^2}$ and $b \in p^2\mathbb{Z}_{p^3}$, where $x = (a, b)$.
So, there are p^2 elements satisfying $px = 0$.

Consequently, there are $p^4 - p^2$ elements x such that $|H| = \langle x \rangle$.

Moreover, the cyclic group of order p^2 has $\phi(p^2) = p^2 - p$ generators,

so the possible $|H| \cong \mathbb{Z}_{p^2}$ are

$$(p^4 - p^2) / (p^2 - p)$$

numbers.

Finally, the number of $|H|$ is $1 + (p^4 - p^2) / (p^2 - p)$.

Note: for a prime p , $\phi(p^n) = p^n - p^{n-1}$.

2020 Fall.

2. For each $n \in \mathbb{N}$, let $a_n = \cos(2\pi/n)$.

a). Prove that a_n is algebraic over $\mathbb{Q}$.

b). Calculate $\deg_{\mathbb{Q}}(a_5)$.

Proof. a). Let $\zeta = \exp(2\pi i/n)$. Then, $\zeta^{-1} = \exp(-2\pi i/n) = \cos(2\pi/n) - i\sin(2\pi/n)$ gives that $a_n = \cos(2\pi/n) = (\zeta + \zeta^{-1})/2$.

Meanwhile, since $\zeta^n = \exp 2\pi i = 1$, so $\zeta^n - 1$ gives ζ is algebraic over $\mathbb{Q}$.

Therefore, ζ^{-1} is also algebraic, thus $a_n = \frac{\zeta + \zeta^{-1}}{2}$ is algebraic over $\mathbb{Q}$.

b). Let $w = \exp(2\pi i/5)$. Since w is the primitive 5th root of unity, $w^4 + w^3 + w^2 + w + 1 = 0$. And, dividing by w^2 then $2a_5 = w + w^{-1}$ implies

$$w^4 + w^3 + w^2 + w + 1 = 0 \Rightarrow w^2 + w^{-2} + w + w^{-1} + 1 = 0$$

$$\Rightarrow 4a_5^2 - 2 + 2a_5 + 1 = 0$$

$$\Rightarrow 4a_5^2 + 2a_5 - 1 = 0.$$

Since $4x^2 + 2x - 1$ is irreducible over $\mathbb{Q}$ by the discriminant $2^2 + 16 = 20$ but $\sqrt{20} \notin \mathbb{Q}$, so $\deg_{\mathbb{Q}}(a_5) = 2$. $\square$

Fall.

a) Notice that De Moivre's formula gives $e^{i \cdot n z} = (\cos n z + i \cdot \sin n z) = (\cos z + i \cdot \sin z)^n$

Therefore

$$\begin{aligned} \cos n \cdot \frac{2\pi}{n} = 1 &= \operatorname{Re} \left(\cos \frac{2\pi}{n} + i \cdot \sin \frac{2\pi}{n} \right)^n \\ &= \operatorname{Re} \left(\sum_{k=0}^n \binom{n}{k} \left(\cos \frac{2\pi}{n} \right)^k \cdot \left(i \cdot \sin \frac{2\pi}{n} \right)^{n-k} \cdot \binom{n}{k} \right) \end{aligned}$$

Meanwhile, $\sin^2(\frac{2\pi}{n}) = 1 - \cos^2(\frac{2\pi}{n})$ and

the real part of $\sum_{k=0}^n \binom{n}{k} \cdot \left(\cos \frac{2\pi}{n} \right)^k \cdot \left(i \cdot \sin \frac{2\pi}{n} \right)^{n-k}$ yield only $n-k$ is even,

since i^{n-k} is real $\Leftrightarrow n-k$, so $\operatorname{Re} \left(\sum_{k=0}^n \binom{n}{k} \cdot \left(\cos \frac{2\pi}{n} \right)^k \cdot \left(i \cdot \sin \frac{2\pi}{n} \right)^{n-k} \right) - 1 = 0$ is evaluated $\cos(\frac{2\pi}{n})$ for some polynomial over $\mathbb{Q}$, so a_n is algebraic over $\mathbb{Q}$.

b). From above argument,

$$\begin{aligned} 1 &= \underbrace{1 \cdot a_5^0}_{k=0} + \underbrace{5 \cdot a_5^1 \cdot (1 - a_5^2)}_{k=1} - \underbrace{10 \cdot a_5^3 \cdot (1 - a_5^2)^2}_{k=3} + \underbrace{a_5^5}_{k=5} \\ &= 5 \cdot a_5 - 5 \cdot a_5^3 - 10 a_5^3 \cdot (1 - 2a_5^2 + a_5^4) + a_5^5 \\ &= 5 \cdot a_5 - 15 a_5^3 + 11 a_5^5 - 10 \cdot a_5^7. \end{aligned}$$

$$\text{So, } 10 a_5^7 - 11 a_5^5 + 15 a_5^3 - 5 a_5 + 1 = 0.$$

2020, Fall

3. Let R be a field. Show that $(x^3 - y^5) \in R[x, y]$ is prime ideal.

Proof. Since $R[x, y]$ is U.F.D., every irreducible element is prime.

Thus, enough to show that $x^3 - y^5$ is irreducible.

Notice that: $R[x, y] = R[y][x]$ by definition, so $x^3 - y^5 \in R[y][x]$.

Since $\gcd(1, y^5) = 1$, $x^3 - y^5$ is primitive, by the Gauss Lemma,

$x^3 - y^5$ is irreducible over $R[y] \Leftrightarrow x^3 - y^5$ is irreducible over $R(y)$.

But, since $x^3 - y^5$ has degree 3, $x^3 - y^5$ is reducible only if it has a linear factor, in other say, $x^3 - y^5$ has a root in $R(y)$.

Suppose that for some $f(y)/g(y) \in R(y)$,

$$\frac{f(y)^3}{g(y)^3} - y^5 = 0.$$

Then, $f(y)^3 = y^5 g(y)^3$, But $\deg f(y)^3 \equiv 0 \pmod{3}$, and

$$\deg y^5 g(y)^3 = 5 + \deg g(y)^3 \equiv 2 \pmod{3}.$$

Therefore, there is no such a root, so $x^3 - y^5$ is irreducible over $R(y)$,
so it is irreducible over $R[x, y]$.

Note:

$$K[x, y] = K[y][x] \xrightarrow{\text{Gauss}} K(y)[x] \xrightarrow{\text{irr. Test.}}$$

Modified

Prove that $\gcd(m,n)=1$ implies $x^m - y^n \in F[x,y]$ is irreducible.

2020. Fall.

4. Let $W \subseteq V = \mathbb{R}^n$ be a subspace. Show that there exists a homogeneous system of linear equations whose solution space is W .

Proof. Let $\mathcal{B} = \{\alpha_1, \dots, \alpha_k\}$ be a basis for W .

Then, we can extend the basis $\mathcal{B}$ to $\overline{\mathcal{B}} = \mathcal{B} \cup \{\alpha_{k+1}, \dots, \alpha_n\}$ for V .

Now, consider the map:

$$f_{\mathcal{B}}: V \rightarrow \mathbb{R}^k; \sum c_i \alpha_i \mapsto C_{\mathcal{B}}$$

Then, $W = \ker f_{k+1} \cap \dots \cap \ker f_n$, so for the standard basis $\mathcal{S}$ for V ,

$$A = \begin{pmatrix} [f_{k+1}]_{\mathcal{S}} \\ \vdots \\ [f_n]_{\mathcal{S}} \end{pmatrix} \text{ satisfies } \ker A = W, \text{ so}$$

([f] $_{\mathcal{S}}$ is row vector representation).

$$A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \text{ is the system we desired.}$$

2021. Spring.

1. Show that no group of 2020 is simple.

Proof $2020 = 2^2 \cdot 5 \cdot 101$. Denote n_p as the number of Sylow p -subgroups.

By the Sylow theorem, $n_{101} \equiv 1 \pmod{101}$ and $n_{101} \mid 20$, so n_{101} must be 1. Therefore, the Sylow 101-subgroup is normal so proved.

Without using Sylow theorem.

By the Cauchy's theorem, G has cyclic subgroup $H \leq G$ of order 101.

Denote $A = \{g \mid g^{-1} \notin G\}$. Then, A is orbit of H from the action

$$G \times \{S \leq G\} \rightarrow \{S \leq G\} : (g, S) \mapsto gSg^{-1}.$$

Since the stabilizer of this action of H is $G_H = N_G(H)$, we obtain

$$\#A = |G : N_G(H)|.$$

And since $H \leq N_G(H)$,

$$|G : N_G(H)| \leq |G : H| = 2020/101 = 20, \text{ so } \#A \leq 20.$$

Now, consider the action

$$H \times A \rightarrow A : (h, K) \mapsto hKh^{-1}.$$

Given $K \in A$, the H -orbit of this action is $\text{Orb}_H(K) = \{hKh^{-1} \mid h \in H\}$ and

$$\#\text{Orb}_H(K) = |H : \text{Stab}_H(K)|, \text{Stab}_H(K) \leq H \Rightarrow |\text{Stab}_H(K)| \in \{1, 101\}.$$

So, $\#\text{Orb}_H(K) \in \{1, 101\}$ and $\text{Orb}_H(K) \leq A$, so $\#\text{Orb}_H(K) = 1$.

Therefore, every element in H normalizes $K \in A$, that is, $H \leq N_G(K)$.

Now, for some $K \in A$, if $H \neq K$ then $HK \leq G$ and

$$|HK| = \frac{|H| \cdot |K|}{|H \cap K|} = 101^2$$

but, since $101^2 \nmid 2020$, so $A = \{1\}$. That is, $H \leq G$ $\square$

2021. Spring

7. Let f be a polynomial given by

$$f(z) = a_0 + a_1 z + \dots + a_d z^d \quad \text{over } \mathbb{C}, \quad a_d \neq 0$$

8m.

Evaluate the value $\lim_{R \rightarrow \infty} \int_{|z|=R} \frac{z \cdot f'(z)}{f(z)} dz$

Proof. By the fundamental theorem of Algebra, f has d -roots $z_1, \dots, z_d \in \mathbb{C}$.

That is, $f(z) = a_d(z - z_1) \dots (z - z_d)$

Moreover, $\frac{f'(z)}{f(z)} = \frac{1}{z - z_1} + \dots + \frac{1}{z - z_d}$, just calculation.

Therefore, for $R > \max\{|z_1|, \dots, |z_d|\}$, by the residue theorem,

$$\frac{1}{2\pi i} \int_{|z|=R} \frac{z \cdot f'(z)}{f(z)} dz = \sum_{n=1}^d \operatorname{Res}\left(\frac{z}{z - z_n}, z_n\right) = \sum_{n=1}^d \lim_{z \rightarrow z_n} z = z_1 + \dots + z_d$$

Consequently, the value is

$$2\pi i \cdot (z_1 + \dots + z_d) = 2\pi i \cdot \left(\frac{-a_{d-1}}{a_d}\right),$$

□

2022 Spring.

8. Define $f: \mathbb{R} \rightarrow \mathbb{R}; x \mapsto \sum_{n=1}^{\infty} \frac{1}{x^2 + n^2}$

Then, f is differentiable on $\mathbb{R}$.

Proof. For each $m \in \mathbb{N}$, define

$$f_m(x) = \sum_{n=1}^m \frac{1}{x^2 + n^2}.$$

At first, $f_m(x) \rightarrow f(x)$ as $m \rightarrow \infty$, because that

$$\sum_{n=2}^m \frac{1}{n^2} < \sum_{n=2}^m \frac{1}{n^2 - n} = \sum_{n=2}^m \frac{1}{n-1} - \frac{1}{n} = 1 - \frac{1}{m} \rightarrow 1 \text{ as } m \rightarrow \infty,$$

so the increasing bounded sequence $f_m(x)$ converges. Now,

$$f'_m(x) = \sum_{n=1}^m -\frac{2x}{(x^2 + n^2)^2}$$

converges uniformly, because

if $|x| \leq 1$, then

$$\left| \frac{-2x}{(x^2 + n^2)^2} \right| \leq \frac{2}{n^4} \text{ and } \sum \frac{1}{n^4} < \infty,$$

if $|x| > 1$, then

$$\left| \frac{-2x}{(x^2 + n^2)^2} \right| \leq 2 \cdot \frac{|x|}{x^2 + n^2} \cdot \frac{1}{x^2 + n^2} \leq 2 \cdot \frac{1}{n^2} \text{ and } \sum \frac{1}{n^2} < \infty$$

therefore, the Weierstrass M-test gives f'_m converges uniformly.

Now, f_m converges at some point and f'_m converges uniformly on $\mathbb{R}$, the limit function f is differentiable. $\square$ sm.

First, for any $x \in \mathbb{R}$, $|x^2 + n^2| \geq n^2$, so

$$\sum_{n=1}^{\infty} \left| \frac{1}{x^2 + n^2} \right| \leq \sum_{n=1}^{\infty} \frac{1}{n^2} < \infty, \text{ because } \sum_{n=2}^m \frac{1}{n^2} < \sum_{n=2}^m \frac{1}{n^2 - n} = \sum_{n=2}^m \frac{1}{n-1} - \frac{1}{n} = 1 - \frac{1}{m} \rightarrow 1,$$

as $m \rightarrow \infty$. Therefore, given ε_0 , for some $N \in \mathbb{N}$, $m > n \geq N$ implies

$$\sup_{x \in \mathbb{R}} \left| \sum_{k=n}^m \frac{1}{x^2 + k^2} \right| \leq \sup_{x \in \mathbb{R}} \sum_{k=n}^m \left| \frac{1}{x^2 + k^2} \right| < \sum_{k=n}^m \frac{1}{k^2} < \varepsilon.$$

Now, define for each $m \in \mathbb{N}$,

$$f_m(x) = \sum_{n=1}^m \frac{1}{x^2 + n^2}.$$

We already shown $f_m \rightarrow f$ uniformly. And,

$$f'_m(x) = \sum_{n=1}^m -\frac{2x}{(x^2 + n^2)^2}$$

converges uniformly, because

2022. Fall.

1. Let G be a group and H and K are normal subgroup of G .

If $H \cap K = \{e\}$, then $H/K \cong H \times K$.

Proof. Define a map

$$\varphi: H \times K \rightarrow H/K: (h, k) \mapsto hK.$$

It is well-defined by the definition of H/K , and it is homomorphism; given $h_1, h_2 \in H$ and $k_1, k_2 \in K$,

$$\varphi((h_1, k_1) \cdot (h_2, k_2)) = \varphi((h_1 h_2, k_1 k_2)) = h_1 h_2 K_1 k_2$$

meanwhile, given $h \in H$ and $k \in K$, $hkh^{-1}k^{-1} \in K$ and $\in H$, because $hkh^{-1} \in K$ and $kh^{-1}k^{-1} \in H$, by the normality. But, $H \cap K = \{e\}$ implies $hkh^{-1}k^{-1} = e$, so $hk = kh$, these commute so,

$$h_1 h_2 k_1 k_2 = h_1 k_1 h_2 k_2 = \varphi((h_1, k_1)) \varphi((h_2, k_2)), \text{ so } \varphi \text{ preserves the operation.}$$

And, $\varphi[H \times K] = H/K$ by just definition, φ is onto. Now, the first isomorphism theorem gives that $H \times K / \ker \varphi \cong H/K$.

$$\text{And, } \ker \varphi = \{(h, k) \in H \times K \mid hK = e\} = \{e\},$$

so $H \times K \cong H \times K / \{e\}$, it proves the assertion. $\square$

2022 Fall.

8. For continuous functions $g: [0,1] \rightarrow \mathbb{R}$ and $h: [-1,1] \rightarrow \mathbb{R}$, define

$$f: [0,1] \rightarrow \mathbb{R}; x \mapsto \int_0^1 h(x-y)g(y) dy$$

For each $n \in \mathbb{N}$, define

$$f_n: [0,1] \rightarrow \mathbb{R}; x \mapsto \frac{1}{n} \cdot \sum_{i=0}^{n-1} h(x - \frac{i}{n}) \cdot g(\frac{i}{n})$$

Prove that $f_n \rightarrow f$ uniformly on $[0,1]$

Proof. Define $l: [0,1] \times [0,1]; (x,y) \mapsto h(x-y) \cdot g(y)$.

Since h and g both are continuous, therefore l is continuous and since $[0,1]^2$ is compact set, so l is continuous uniformly.

Given $\epsilon > 0$, there exists $\delta > 0$ such that for any $(x_1, y_1), (x_2, y_2) \in [0,1]^2$,

$$\|(x_1, y_1) - (x_2, y_2)\| < \delta \Rightarrow \|l(x_1, y_1) - l(x_2, y_2)\| < \epsilon.$$

Now, given $x \in [0,1]$, and $n > \frac{1}{\delta}$, ($n > \frac{1}{\delta}$ gives $\frac{1}{n} < \delta$)

$$\begin{aligned} |f(x) - f_n(x)| &= \left| \int_0^1 l(x, y) dy - f_n(x) \right| \\ &= \left| \sum_{i=0}^{n-1} \int_{\frac{i}{n}}^{\frac{i+1}{n}} l(x, y) dy - \frac{1}{n} \cdot \sum_{i=0}^{n-1} l(x, \frac{i}{n}) \right| \\ &= \left| \sum_{i=0}^{n-1} \int_{\frac{i}{n}}^{\frac{i+1}{n}} l(x, y) dy - \sum_{i=1}^{n-1} \int_{\frac{i}{n}}^{\frac{i+1}{n}} l(x, \frac{i}{n}) dy \right| \\ &\leq \sum_{i=0}^{n-1} \int_{\frac{i}{n}}^{\frac{i+1}{n}} |l(x, y) - l(x, \frac{i}{n})| dy \\ &< \sum_{i=0}^{n-1} \int_{\frac{i}{n}}^{\frac{i+1}{n}} \epsilon dy = \epsilon. \end{aligned}$$

Therefore, $n > \frac{1}{\delta} \Rightarrow \sup_{x \in [0,1]} |f_n(x) - f(x)| < \epsilon$, So, proved. $\square$

12022. Fall.

18. Let S be the subset of $\mathbb{R}^3$ given by $\{(x, y, z) \mid z^2 = x^2 + y^2, z \geq 0\}$.

Show that S is not regular surface.

Proof. Suppose that S is a regular surface. Then, for $(0, 0, 0) \in S$, there exists a neighbor V containing $(0, 0, 0)$ such that $V \cap S$ is a graph of

$$z = f(x, y) \quad \text{or} \quad y = g(x, z) \quad \text{or} \quad z = h(y, z)$$

where f, g, h are differentiable. Since projection of S into xz , yz -plane are not 1-1, g, h cannot be such a map.

Meanwhile, $f(x, y) = \sqrt{x^2 + y^2}$ at locally $(0, 0, 0)$, but f is not differentiable at $(x, y) = (0, 0)$, so it cannot be possible. Therefore S is not regular.

(More specifically, $(0, \pm \epsilon, \sqrt{1 - \epsilon^2}) \in S$ for any $\epsilon > 0$, so there is no 1-1 yz -projection, similarly xz .)

2023. Spring

4. Let $V = \{id, (12)(34), (13)(24), (14)(23)\} \subseteq S_4$, 57,

a). Show that V is normal subgroup.

b). $S_4/V \cong S_3$,

a). Since conjugate cannot change the order of elements, and cannot change the parity: since the sign function is Homomorphism,

$$\text{sgn}(\sigma\tau\sigma^{-1}) = \text{sgn}(\sigma) \cdot \text{sgn}(\sigma^{-1}) \cdot \text{sgn}(\tau) = \text{sgn}(\tau), \text{ for any } \sigma, \tau \in S_n.$$

Now, the fact that V contains all order two even permutations in S_4 , so every conjugate of V is contained in V , so normal. $\downarrow$ 5m

b). Consider S_3 is a subgroup of S_4 , the number 4 fixed. Then,

$$\mathcal{Q}: S_3 \rightarrow S_4/V: \sigma \mapsto \sigma V$$

is well-defined, and given $\sigma_1, \sigma_2 \in S_3 \leq S_4$,

$$\mathcal{Q}(\sigma_1\sigma_2) = \sigma_1\sigma_2 V = \sigma_1 V \sigma_2 V = \mathcal{Q}(\sigma_1) \mathcal{Q}(\sigma_2)$$

so it Homomorphism. And, since all elements in S_3 fixes 4, $S_3 \cap V = \{id\}$, so $\ker \mathcal{Q} = \{id\}$. Now, since $|S_3| = 6 = |\ker \mathcal{Q}| \cdot |\text{Im } \mathcal{Q}|$, $|\text{Im } \mathcal{Q}| = 6$, so $\text{Im } \mathcal{Q} = S_4/V$, since $|S_4/V| = 6$, so $\mathcal{Q}$ is onto.

Now, the first isomorphism theorem gives $S_3 \cong S_4/V$. $\square$

2023 Spring

5. Let $I = (2, x^2 + x + 1) \subseteq \mathbb{Z}[x]$ be an ideal

show that I is maximal ideal and give a multiplication table for $\mathbb{Z}[x]/I$.

Proof. Define a map

$$\pi: \mathbb{Z}[x] \rightarrow \mathbb{Z}/2\mathbb{Z}[x]; \sum_{n=0}^{\infty} a_n x^n \mapsto \sum_{n=0}^{\infty} (a_n \bmod 2) x^n$$

Then, π is an onto Homomorphism, ($\pi(0) = 0$, $\pi(1) = 1$) with $\ker \pi = 2\mathbb{Z}[x] = (2)$, so $\mathbb{Z}[x]/\ker \pi = \mathbb{Z}[x]/2\mathbb{Z}[x] \cong (\mathbb{Z}/2\mathbb{Z})[x]$.

Meanwhile, $x^2 + x + 1$ is irreducible over $\mathbb{Z}/2\mathbb{Z}$, because it is of degree 2 and $0^2 + 0 + 1 = 1 \neq 0$, $1^2 + 1 + 1 = 1 \neq 0$ in $\mathbb{Z}/2\mathbb{Z}$.

Now, Consider the natural Homomorphism

$$\phi: \mathbb{Z}/2\mathbb{Z}[x] \rightarrow (\mathbb{Z}/2\mathbb{Z}[x])/(x^2 + x + 1); p(x) \mapsto p(x) + (x^2 + x + 1)$$

It is trivially onto, so $\phi \circ \pi$ is onto Homomorphism.

Moreover,

$$\ker(\phi \circ \pi) = (\phi \circ \pi)^{-1}[\{0\}] = \pi^{-1}[\phi^{-1}[\{0\}]] = \pi^{-1}[\ker \phi] = \pi^{-1}[(x^2 + x + 1)]$$

Now,

$$f \in \pi^{-1}[(x^2 + x + 1)] \Leftrightarrow \pi(f) \in (x^2 + x + 1)$$

$$\Leftrightarrow \pi(f) = (x^2 + x + 1) \cdot \pi(g) \text{ for some } g(x) \in \mathbb{Z}[x]$$

$$\Leftrightarrow \pi(f - (x^2 + x + 1)g) = 0$$

$$\Leftrightarrow f - (x^2 + x + 1)g \in (2)$$

$$\Leftrightarrow f \in (2) + (x^2 + x + 1) = (2, x^2 + x + 1).$$

So, $\mathbb{Z}[x]/\ker(\phi \circ \pi) = \mathbb{Z}[x]/(2, x^2 + x + 1) \cong (\mathbb{Z}/2\mathbb{Z})[x]/(x^2 + x + 1)$.

And, since $\mathbb{Z}/2\mathbb{Z}$ is field and $x^2 + x + 1$ is irreducible over $\mathbb{Z}/2\mathbb{Z}$, so $(\mathbb{Z}/2\mathbb{Z})[x]/(x^2 + x + 1)$ is a field, so, the quotient is field if and only if $(x^2 + x + 1)$ is maximal, so proved.

Since $\mathbb{Z}[x]/I \cong (\mathbb{Z}/2\mathbb{Z})[x]/(x^2 + x + 1)$, Let $\alpha = x + (x^2 + x + 1)$.

Then $\alpha^2 = x^2 + (x^2 + x + 1) = x + 1 + (x^2 + x + 1) = \alpha + 1$, so

	1	α
1	1	α
α	α	$\alpha + 1$

is the table,

Generalized in the next page.

Let $I = (a, f) \subseteq R[x]$ be an ideal generated by $a, f \in R[x]$.

Consider the diagram:

$$\begin{array}{ccccc} R[x] & \xrightarrow{\mathcal{Q}} & (R/aR)[x] & \xrightarrow{\pi_f} & (R/aR)[x]/(\overline{f}) \\ & \searrow \pi & \cong \nearrow & & \\ & & R[x]/aR[x] & & \end{array}$$

$$\begin{array}{ll} R[x] \rightarrow R[x]/aR[x] & R[x]/aR[x] \rightarrow (R/aR)[x] \\ p(x) \mapsto p(x) + aR[x] & p(x) + aR[x] \mapsto \overline{p(x)}, \text{ where } \overline{\sum a_n \cdot x^n} = \sum (a_n + aR) \cdot x^n. \end{array}$$

Onto Homomorphism.

Isomorphism.

$\Rightarrow \mathcal{Q}: R[x] \rightarrow (R/aR)[x]$ is Onto Homomorphism, so is $\pi_f \circ \mathcal{Q}$ onto Homomorphism.

Moreover,

$$\begin{aligned} \ker(\pi_f \circ \mathcal{Q}) &= (\pi_f \circ \mathcal{Q})^{-1}[\{0\}] = \mathcal{Q}^{-1}[\ker \pi_f] \\ &= \{p(x) \in R[x] \mid \overline{p(x)} \in \ker \pi_f\} = \{p(x) \in R[x] \mid \overline{p(x)} \in (\overline{f_{00}})\} \\ &\stackrel{(*)}{=} \{p(x) \in aR[x] + f_{00}R[x]\} = (a) + (f) = (a, f) = I. \end{aligned}$$

Assume $a \in R$ and $(a) \subseteq R$ is maximal, and $\overline{f} \in (R/aR)[x]$ is irreducible.

Then, R/aR is a field, and therefore $(R/aR)[x]$ is P.I.D.

So, $(\overline{f})$ is maximal, it implies that $(R/aR)[x]/(\overline{f})$ is a field.

Now, by the first-isomorphism theorem,

$$R[x]/(aR[x] + fR[x]) \cong (R/aR)[x]/\overline{f} \cdot (R/aR)[x]$$

Note: (*) is obtained by: Given $p(x) \in R[x]$,

$$\overline{p(x)} \in (\overline{f_{00}}) \Leftrightarrow \overline{p(x)} = \overline{g(x)} \cdot \overline{f(x)} \text{ for some } g(x) \in R[x]$$

$$\Leftrightarrow p(x) - g(x)f(x) \in aR[x] \text{ for some } g(x) \in R[x]$$

2023, Spring.

7. Let $f: \mathbb{C} \rightarrow \mathbb{C}$ be an entire satisfying 10m

$$U^2(z) - 2 \cdot V^2(z) = 1 \text{ for all } z \in \mathbb{C}$$

Prove that f is constant.

Proof. Observe that ~~differentiating gives~~ that

$$2U U_x - 2V V_x = 0 \Rightarrow \cancel{U U_x} = V V_x = -V \cdot U_y$$

Observe that: $f(z) = 0 \Leftrightarrow U(z) = V(z) = 0$

But, $U^2 - 2V^2 = 1$, so $f(z) \neq 0$ for all $z \in \mathbb{C}$. Now,

$$\frac{1}{f(z)} = \frac{1}{U^2(z) + V^2(z)} = \frac{1}{1 + 3V^2(z)} \leq 1,$$

therefore by the Liouville's theorem, the bounded analytic function $\frac{1}{f^2}$ is constant, so f is also constant function. $\square$

$$\text{Or, } |f^2| = U^2 + V^2 = 1 + 3V^2 \Rightarrow \frac{1}{f} \leq \frac{1}{\sqrt{1+3V^2}} \leq 1.$$

2023 Spring

10. Let $f: [0, \infty) \rightarrow [0, \infty)$ be a continuous function satisfying

$$\int_0^{\infty} f(x) dx < \infty.$$

a). Prove or disprove that $\lim_{x \rightarrow \infty} f(x) = 0$.

b). If f is continuous uniformly then $\lim_{x \rightarrow \infty} f(x) = 0$? Prove or disprove.
Proof.

a). Define a map on $[0, \infty)$

$$f_n(x) = \begin{cases} 0 & \text{otherwise} \\ n^2(x-n) & \text{if } x \in [n, n + \frac{1}{n^2}] \\ n^2(n + \frac{2}{n^2} - x) & \text{if } x \in [n + \frac{1}{n^2}, n + \frac{2}{n^2}] \end{cases}$$

Then $\int_0^{\infty} f_n = \frac{1}{n}$ and f_n is continuous, for each $n \in \mathbb{N}$.

Let $f = f_1 + f_2 + \dots + f_n + \dots$. Then f is continuous and

$$\int_0^{\infty} f = \sum_{n=1}^{\infty} \frac{1}{n^2} < \infty.$$

(The continuity is obtained just investigate each $[n, n + \frac{2}{n^2}]$)

But, for $\epsilon = \frac{1}{2}$, for any $\delta > 0$, there is $x > \delta$ such that $|f(x) - 0| = 1 \geq \frac{1}{2}$.
So $\lim_{x \rightarrow \infty} f(x) \neq 0$.

b). Suppose $\lim_{x \rightarrow \infty} f(x) = 0$ is false but f is continuous uniformly.

Let $\epsilon = 2$. Then, for any $n \in \mathbb{N}$, there is $x_n > n$ such that $f(x_n) \geq 2$.

Since f is uniformly continuous, take $\delta > 0$ such that $|x - y| < \delta$ implies $|f(x) - f(y)| < 1$. Then, for each $n \in \mathbb{N}$,

$$\int_{x_n}^{x_n + \delta} f(x) dx \geq \delta \cdot 1$$

So, $\int_0^{\infty} f \geq \sum_{n=1}^N \int_{x_n}^{x_n + \delta} f \geq \delta \cdot N$ diverges, So contradicts.

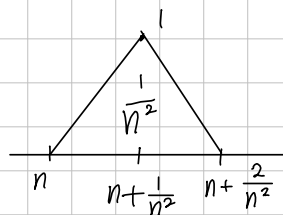
Thus $\lim_{x \rightarrow \infty} f = 0$. $\square$

Revised in the Next.

Let $f: [0, \infty) \rightarrow \mathbb{R}$ be non-negative valued continuous function, does not Converges to Zero, but the 'improper integral' Converges.

Find and Convince Concretely, and change the Condition as f Continuous Uniformly.

Define $f_n(x) = \begin{cases} n^2(x-n) & \text{if } x \in [n, n + \frac{1}{n^2}] \\ n^2(n + \frac{2}{n^2} - x) & \text{if } x \in [n + \frac{1}{n^2}, n + \frac{2}{n^2}] \\ 0 & \text{otherwise.} \end{cases}$



Then, f_n is continuous, with $\int_0^\infty f_n = \frac{1}{n^2}$.

Now, let $f = \sum_{n=2}^\infty f_n$. Since the interval $[n, n + \frac{2}{n^2}]$ are disjoint mutually, $n \geq 2$, so the f is well-defined, but it does not uniformly Converges, because

$$\sup_{x \geq 0} \left| f(x) - \sum_{n=2}^N f_n(x) \right| = 1 \text{ for any } N \in \mathbb{N}.$$

Nevertheless, $\int_0^\infty f = \int_0^\infty \sum_{n=2}^\infty f_n = \sum_{n=2}^\infty \int_0^\infty f_n = \sum_{n=2}^\infty \frac{1}{n^2}$, the integral Converges.

But, for each $n \in \mathbb{N}$, $f(n + \frac{1}{n^2}) = 1$, so $f(x)$ does not tends to Zero as $x \rightarrow \infty$.

Assume that $f(x)$ is Continuous Uniformly

If f does not Converges to Zero, then there is $\epsilon > 0$ such that for each $R > 0$, there is $x \geq R$ such that $f(x) \geq \epsilon$.

Now, let $\delta > 0$ such that $|x - y| < \delta \Rightarrow |f(x) - f(y)| < \frac{\epsilon}{2}$.

And, let $\{x_n\}$ be a Positive Sequence such that for all $n \in \mathbb{N}$,

$$x_{n+1} > x_n + 2\delta \text{ and } f(x_n) \geq \epsilon.$$

$$\text{Then, } \int_0^{x_n + \delta} f \geq \sum_{k=2}^n \frac{\epsilon}{2} \cdot 2\delta = n\epsilon\delta \rightarrow \infty \text{ as } n \rightarrow \infty,$$

so it Contradicts to $\int_0^\infty f < \infty$. $\square$

2024, Spring.

1. Let $A, B \in M_{m \times n}(F)$. Show that $\text{rank}(A+B) \leq \text{rank } A + \text{rank } B$. [5m]

Proof. By definition,

$$\text{rank}(A+B) = \dim \text{Im}(A+B) \text{ and } \text{rank } A = \dim \text{Im } A, \text{ rank } B = \dim \text{Im } B.$$

Given $\alpha \in \text{Im}(A+B)$, $\alpha = (A+B)\beta = A\beta + B\beta \in \text{Im } A + \text{Im } B$ for some $\beta \in F^n$, therefore $\text{Im}(A+B) \subseteq \text{Im } A + \text{Im } B$. Now,

$$\text{rank}(A+B) \leq \dim(\text{Im } A + \text{Im } B) \leq \dim \text{Im } A + \dim \text{Im } B = \text{rank } A + \text{rank } B.$$

2. Suppose that P and Q are positive definite such that $P^2 = Q^2$. Show $P = Q$.

Proof. Since P is Hermitian, there exists an orthonormal basis $\mathcal{B} = \{\alpha_1, \dots, \alpha_n\}$ such that $P\alpha_i = C_i\alpha_i$ for some $C_i > 0$, for each i .

Let $1 \leq i \leq n$ be fixed. Then, by the assumption,

[5m]

$$P^2\alpha_i = C_i^2\alpha_i = Q^2\alpha_i \Rightarrow (Q^2 - C_i^2I)\alpha_i = (Q - C_iI)(Q + C_iI)\alpha_i = 0.$$

Since Q is positive definite, $(Q - C_iI)\alpha_i = 0$, because: if it is not zero, then $(Q - C_iI)(Q + C_iI)\alpha_i = (Q + C_iI)(Q - C_iI)\alpha_i = 0$,

so $-C_i$ is negative eigenvalue of Q , it contradicts to Q is positive definite. Therefore, for each $1 \leq i \leq n$,

$$P\alpha_i = C_i\alpha_i = Q\alpha_i,$$

and since $\mathcal{B}$ is basis, therefore $P = Q$. $\square$

2024, Spring

3. Let $T: \mathbb{R}^5 \rightarrow \mathbb{R}^5; x \mapsto Ax$ where

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

a). There exists T -invariant subspace $V \subseteq \mathbb{R}^5$ s.t. $\dim V = 4$?

b). "

s.t. $\dim V = 2$, $T|_V$ is diagonalizable?

c). "

$\dim V = 3$, "

Proof.

a). Let $V = \langle e_1, e_2, e_3, e_4 \rangle$. Then,

$$Ae_1 = e_1, Ae_2 = e_1 + e_2, Ae_3 = e_3, Ae_4 = e_3 + e_4$$

are all contained in V , so $T[V] \subseteq V$. (clearly $\dim V = 4$)

b). Let $V = \langle e_1, e_3 \rangle$. Then $Ae_1 = e_1$, $Ae_3 = e_3$ are contained in V , and

$$[T|_V]_{\{e_1, e_3\}} = I_2. \text{ So } T|_V \text{ is diagonalizable.}$$

c). Observe characteristic polynomial of T is $(t-1)^5$.

If such a subspace V exists, then the characteristic polynomial of $T|_V$ is

$(t-1)^3$. Since $T|_V$ is diagonalizable if and only the minimal polynomial

splits and separable, thus the minimal polynomial of $T|_V$ must be $(t-1)$.

That is, $T|_V = I_V$ but it implies that $\ker(T-I)$ has dimension more than or equal 3, it contradicts to

$$\dim \ker(A-I) = \dim \ker \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} = 2.$$

2024, Spring.

5. Let K be a field, and $L_1, L_2 \subseteq \overline{K}$ be finite extensions of K . 41

a). Prove $L = L_1 L_2$ is finite extension.

b). If L_1 and L_2 are Galois over K , then L/K is Galois.

a). Since L_1 and L_2 both are finite, denote

$$L_1 = K(\alpha_1, \dots, \alpha_n) \quad \text{and} \quad L_2 = K(\beta_1, \dots, \beta_r) \quad \text{where } \alpha_i, \beta_j \in \overline{K} \text{ are algebraic over } K.$$

$$\text{Now, } L = L_1 L_2 = \bigcap_{\substack{E \subseteq \overline{K} \\ L_1, L_2 \subseteq E}} E = K(\alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_r)$$

is finitely generated by algebraic elements α_i, β_j , so L is finite. 14m.

b). Since L_1 and L_2 are splitting field of K , let f_1 and f_2 be polynomials over K such that L_i is the splitting field of f_i , $i=1,2$. It is possible, because L_1, L_2 both are finite extension.

Now let E be the splitting field of f_1, f_2 . Then, $E \subseteq L_1 L_2$, because E is the smallest field containing all roots of f_1, f_2 , $L_1 L_2 \subseteq E$ because E contains all roots of f_1 and f_2 . So, $L = E$. Therefore L is splitting field over K . 17m.

Since L_1 and L_2 both are Galois, for some separable f_1 and f_2 over K , L_1 and L_2 are splitting field of f_1 and f_2 respectively.

$L_1 L_2$ is f_1, f_2 splitting field $\Rightarrow OK$

즉 f_1, f_2 is separable 이란 보장 X.

중복 linear factor 소거 $\Rightarrow f_1, f_2$ 가 irreducible over K 인가?

Revised in the Algebra/Field Theory/

2024 Spring.

7. Let $f: (-1, 1) \rightarrow \mathbb{R}$ be a differentiable function satisfying $f(0) = 0$ and

$$\text{let } g: (-1, 1) \rightarrow \mathbb{R}: \begin{cases} f(x)/x & \text{if } x \neq 0 \\ f'(0) & \text{if } x = 0. \end{cases}$$

a). If $f \in C^2$, then $g \in C^1$ (prove)

b). If $f \in C^1$ but $\notin C^2$, then $g \in C^1$ may not hold.

Proof. Notice that g is continuous, since

$$\lim_{x \rightarrow 0} g(x) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = f'(0).$$

a) For $x \neq 0$,

$$g'(x) = \left(\frac{f(x)}{x} \right)' = \frac{x f'(x) - f(x)}{x^2} \text{ exists, and}$$

1/22/24 spring.

8. Let $C_b(\mathbb{R})$ be the set of all real-valued bounded continuous functions on $\mathbb{R}$. For two functions $f, g \in C_b(\mathbb{R})$, define

$$d(f, g) := \sup_{x \in \mathbb{R}} |f(x) - g(x)|.$$

Let $\{f_n\}_{n=1}^{\infty}$ be a sequence in $C_b(\mathbb{R})$ such that:

$\forall \varepsilon > 0, \exists N \in \mathbb{N}$ such that $m, n \geq N \Rightarrow d(f_m, f_n) < \varepsilon$.

a). Prove that there exists $f \in C_b(\mathbb{R})$ such that $\lim_{n \rightarrow \infty} d(f, f_n) = 0$

b). Prove that $\lim_{n \rightarrow \infty} \left(\sup_{x \in \mathbb{R}} |f_n(x)| \right) = \sup_{x \in \mathbb{R}} |f(x)|$.

[Proof. For each $x \in \mathbb{R}$, a sequence $\{f_n(x)\}$ in $\mathbb{R}$ is Cauchy sequence by hypothesis. Since $\mathbb{R}$ is complete, $\lim_{n \rightarrow \infty} f_n(x)$ exists, put this $f(x) \in \mathbb{R}$.

a).

Since $x \in \mathbb{R}$ was arbitrary, $f(x)$ is well-defined on $\mathbb{R}$, but still not necessarily $f \in C_b(\mathbb{R})$, and $d(f, f_n) \rightarrow 0$.

Given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $m > n \geq N$ implies that $\sup_{x \in \mathbb{R}} |f_m(x) - f_n(x)| \leq \varepsilon$.

So, for any $x \in \mathbb{R}$, $|f_m(x) - f_n(x)| \leq \varepsilon$. Letting $m \rightarrow \infty$, we obtain $|f(x) - f_n(x)| \leq \varepsilon$.

Since x was arbitrary, $\sup_{x \in \mathbb{R}} |f(x) - f_n(x)| \leq \varepsilon$, for $n \geq N$.

Therefore, the triangle inequality gives $|f(x)| \leq |f_n(x)| + \varepsilon$ for all $x \in \mathbb{R}$

and since f_n was bounded, f is bounded. And, the argument above

proves $\sup_{x \in \mathbb{R}} |f - f_n| \rightarrow 0$ as $n \rightarrow \infty$, so since f_n is continuous and

$f_n \rightarrow f$ uniformly, so f is also continuous, so $f \in C_b(\mathbb{R})$ and a) is proved.

b). Since $\left| \sup_{x \in \mathbb{R}} |f_n(x)| - \sup_{x \in \mathbb{R}} |f(x)| \right| \leq \sup_{x \in \mathbb{R}} |f_n(x) - f(x)| \rightarrow 0$ as $n \rightarrow \infty$

Prove directly.

2024, Fall.

1. Let V be a finite-dimensional vector space over a field F .
Let $T: V \rightarrow V$ be a linear operator such that $T^2 = T$. Prove

η_m

a). $V = \ker T \oplus \operatorname{Im} T$

b). T is diagonalizable.

Proof. a). Given $\alpha \in V$, $\alpha = \alpha - T(\alpha) + T(\alpha)$, $T(\alpha) \in \operatorname{Im} T$ is trivial,
and $T(\alpha - T(\alpha)) = T(\alpha) - T^2(\alpha) = (T - T^2)(\alpha) = 0$, so $\alpha - T(\alpha) \in \ker T$.

Since α was arbitrary, $V = \ker T + \operatorname{Im} T$.

Moreover, let $\alpha \in \ker T \cap \operatorname{Im} T$. Then, $T(\alpha) = 0$ and $\alpha = T(\beta)$ for some $\beta \in V$. Now, $\alpha = T(\beta) = T^2(\beta) = T(T(\beta)) = T(\alpha) = 0$, so $\ker T \cap \operatorname{Im} T = \{0\}$,
it proves $V = \ker T \oplus \operatorname{Im} T$.

b). Since $T^2 - T = 0$, the minimal polynomial m of T divides $x^2 - x$.
If $m = x$ or $x - 1$, then $T = 0$ or $T = \operatorname{Id}$, already diagonal.
If $m = x(x-1)$, then it splits over the given field and separable,
so T is diagonalizable. $\square$

Note: Naturally, $T|_{\operatorname{Im} T} = \operatorname{Id}$, so a) implies b).

2024 Fall.

8. Solve the followings: $x \in \mathbb{R}$,

a). $x''(t) - 2x'(t) + x(t) = 0$

b). $x''(t) - 2x'(t) + x(t) = te^t + 4$.

Proof.

a). Since the characteristic polynomial of the equation is

$$r^2 - 2r + 1 = (r-1)^2,$$

the general solution is

$$x_1(t) = C_1 e^t + C_2 \cdot t e^t \quad \text{where } C_1, C_2 \in \mathbb{R}.$$

b). Calculate the Wronskian:

$$W(e^t, t e^t) = \begin{vmatrix} e^t & t e^t \\ e^t & e^t + t e^t \end{vmatrix} = e^{2t} + t e^{2t} - t e^{2t} = e^{2t}.$$

Therefore, the particular solution is

$$\begin{aligned} Y(t) &= \int_{t_0}^t \begin{vmatrix} e^u & e^t \\ u e^u & t e^t \end{vmatrix} \cdot b(u) / W(e^u, u e^u) du \\ &= \int_{t_0}^t (e^u t e^t - u e^u e^t) \cdot (u e^u + 4) / e^{2u} du \\ &= \int_{t_0}^t (u e^{2u} t e^t + 4 e^u t e^t - u^2 e^{2u} e^t - 4 u e^u e^t) / e^{2u} du \\ &= \int_{t_0}^t u \cdot t e^t + 4 \cdot t e^t \cdot e^{-u} - u^2 \cdot e^t - 4 e^t \cdot u e^{-u} du \end{aligned}$$

Now, $x_2 + Y$ is the general solution.

2024, Fall,

1. For a fixed $\xi \in \mathbb{R}$, let

$$f(z) = \frac{e^{-2\pi i z \xi}}{\cosh(\pi z)} \quad (z \in \mathbb{C}).$$

a). Show that $\alpha = i/2$ and $\beta = 3i/2$ are simple poles of f .

b). Compute the residues of f at α and β .

c). Evaluate $\int_{-\infty}^{\infty} \frac{e^{-2\pi i x \xi}}{\cosh(\pi x)} dx$.

Proof. a). Notice that $e^{-2\pi i z \xi}$ is entire and has no zero, we need to investigate pole of $1/\cosh(\pi z)$.

Since $\cosh(\pi z) = 0$ if $z_0 = \alpha$ or β , and

$$\lim_{z \rightarrow z_0} \frac{\cosh(\pi z) - \cosh(\pi z_0)}{z - z_0} = \sinh(\pi z_0) \cdot \pi \neq 0, \quad \text{so}$$

$$\lim_{z \rightarrow z_0} (z - z_0) \cdot \frac{1}{\cosh(\pi z)} = \frac{1}{\pi \sinh(\pi z_0)} \text{ exists, so } \alpha \text{ and } \beta \text{ are simple poles.}$$

b). Since α and β both are simple pole,

$$\text{Res}(f, \alpha) = \lim_{z \rightarrow \alpha} (z - \alpha) \cdot f(z) = e^{-2\pi i \alpha \xi} \cdot \frac{1}{\pi \cdot \sinh(\pi \alpha)} = \frac{e^{\pi \xi}}{\pi \cdot \sinh(\pi i/2)} = \frac{e^{\pi \xi}}{\pi i}.$$

$$\text{Res}(f, \beta) = e^{-2\pi i \beta \xi} \cdot \frac{1}{\pi \cdot \sinh(\pi \beta)} = \frac{e^{-3\pi \xi}}{\pi \cdot (-i)} = -\frac{e^{-3\pi \xi}}{\pi i}.$$

c). Let C_R be the rectangle $C_R = C_1 * C_2 * C_3 * C_4$ along four line segments $-R$ to R , R to $R+i$, $R+i$ to $-R+i$, $-R+i$ to $-R$.

And, observe that

$$f(x+i) = \frac{e^{-2\pi i x \xi + 2\pi \xi}}{\cosh(\pi x + \pi i)} = e^{2\pi \xi} \cdot -f(x)$$

$$\text{Therefore, } \int_{C_1 + C_3} f = \left(\int_{-R}^R f(x) dx \right) \cdot (1 + e^{2\pi \xi}).$$

And, since $|f(R+it)| = \left| \frac{e^{-2\pi i R \xi + 2\pi t \xi}}{\cosh(\pi R + \pi it)} \right| \rightarrow 0$ as $|\cosh(\pi R + \pi it)| \rightarrow \infty$ as $R \rightarrow \infty$.

Therefore, $\int_{C_2} f, \int_{C_4} f \rightarrow 0$, so, by the Residue theorem,

$$\int_{-\infty}^{\infty} f(x) dx = \frac{1}{1 + e^{2\pi \xi}} \cdot \lim_{R \rightarrow \infty} \int_{C_R} f = \frac{1}{1 + e^{2\pi \xi}} \cdot 2e^{\pi \xi}.$$

2025 Spring.

13. Let D^2 be a closed unit disc in $\mathbb{R}^2$ centered at the origin.

a). Show that the quotient space $\mathbb{R}^2/D^2$ is homeomorphic to $\mathbb{R}^2$.

b). Let $U = \text{int } D$. Determine $\mathbb{R}^2/U \cong \mathbb{R}^2$.

Proof. Let $\sim$ be a relation defined on $\mathbb{R}^2$ as:

$$(x_1, y_1) \sim (x_2, y_2) \Leftrightarrow \{(x_1, y_1), (x_2, y_2)\} \subseteq D^2.$$

Then, it is clearly an equivalence relation, and $\mathbb{R}^2/\sim = \mathbb{R}^2/D^2$, by definition.

a). Define a map

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}^2; (x, y) \mapsto$$

Then, given $(x_1, y_1), (x_2, y_2) \in \mathbb{R}^2$,

$$(x_1, y_1) \sim (x_2, y_2) \Leftrightarrow$$

2025. Spring.

14. For $a, b, c > 0$, Consider the ellipsoid

$$E_{a,b,c} = \left\{ (x, y, z) \in \mathbb{R}^3 \mid \left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 + \left(\frac{z}{c}\right)^2 = 1 \right\}.$$

- a). If $a=b$, then $E_{a,a,c}$ is a surface of revolution (Prüferl.)
- b) Give a parametrization of $E_{a,a,c}$. Compute the first fundamental form of this parametrization and the Christoffel symbol.
- c). Show that the Curve $E_{a,b,c} \cap \{x=0\}$ is the image of a geodesic.

a). Let

$$X: (0, 2\pi) \times (0, 2\pi) \rightarrow \mathbb{R}^3; (u, v) \mapsto (a \cdot \cos v \cdot \cos u, a \cdot \cos v \cdot \sin u, c \cdot \sin v).$$

2025, Fall

1. a). Show that $\phi: P_n(\mathbb{R}) \rightarrow \mathbb{R}: p(t) \mapsto p'(0)$ is linear functional.

8m

b). Find a dual basis corresponding to $\{1, t, \dots, t^n\}$ of $P_n(\mathbb{R})$.

Proof. Let $f(x) = \sum_{i=0}^n a_i x^i$ and $g(x) = \sum_{i=0}^n b_i x^i$ and $C \in \mathbb{R}$ be given.

Then, $\phi(Cf + g) = (Cf + g)'|_{t=0} = C \cdot f'|_{t=0} + g'|_{t=0} = C \cdot \phi(f) + \phi(g)$

by the linearity of differentiation. This shows the a),

To show b), let

$$\mathcal{B}^* := \left\{ \frac{1}{k!} \cdot \phi_k: P_n(\mathbb{R}) \rightarrow \mathbb{R}: p(t) \mapsto p^{(k)}(0) \mid k=0, 1, \dots, n \right\}$$

Then, for each $k=0, 1, \dots, n$,

$$\frac{1}{k!} \cdot \phi_k(t^k) = \frac{k!}{k!} \cdot 1 = 1, \text{ and } \frac{1}{k!} \cdot \phi_k(t^r) = 0 \text{ if } r \neq k.$$

Similarly a), the elements in $\mathcal{B}^*$ are all linear functional, so the $\mathcal{B}^*$ is dual basis.

2025. Fall

2. Let $A \in M_{\text{max}}(1\mathbb{R})$.

15m: A^n also 연산, $n=2$ 만 보이기 시작

a). For $n \in \mathbb{N}$, $\text{rank } A^n \leq \text{rank } A$.

b). Let $L_A: \mathbb{R}^m \rightarrow \mathbb{R}^m, x \mapsto Ax$.

Show that for $n \geq 2$,

$$\text{rank } A^n = \text{rank } A \text{ implies } \ker L_A \cap \text{Im } L_A = \{0\}.$$

Sol) a). Given $x \in \text{Im } A^n$, $x = A^n y$ for some $y \in \mathbb{R}^m$.

So, $x = A^n y = A^{n-1} A y \in \text{Im } A$, for $n \geq 2$. Thus $\text{Im } A^n \subseteq \text{Im } A$, it follows $\text{rank } A^n \leq \text{rank } A$. If $n=1$, clear.

b). Given $x \in \ker A$, $A^n x = A^{n-1} A x = A^{n-1} 0 = 0$, so $x \in \ker A^n$, so $\ker A \subseteq \ker A^n$. Since $\text{rank } A^n = \text{rank } A$, the dimension theorem gives $\text{nullity } A^n = \text{nullity } A$.

Therefore, since $\mathbb{R}^m$ is finite dimensional, so is $\ker A^n$, thus $\ker A = \ker A^n$.

Now, let $x \in \ker L_A \cap \text{Im } L_A$ be given. Then, $Ax = 0$ and $x = Ay$ for some $y \in \mathbb{R}^m$.

Then, $A^{n-1} x = A^n y = 0$, because $Ax = 0$, so $A^{n-2} Ax = 0$, and this implies $y \in \ker A^n$.

Since $\ker A^n = \ker A$, $Ay = 0$, so $x = Ay = 0$, so proved. $\square$

2025. Fall

3. Find all $a, b, c \in \mathbb{R}$ such that

6m

$$A = \begin{bmatrix} 2 & 0 & 0 \\ a & 2 & 0 \\ b & c & -1 \end{bmatrix} \text{ is diagonalizable.}$$

Sol. Since the characteristic polynomial of A is regardless of a, b, c ,

$$\chi_A(t) = (t-2)^2(t+1).$$

Since a Matrix is diagonalizable if and only if the minimal polynomial splits and separable, and the minimal polynomial divides the characteristic polynomial, the A is diagonalizable if and only if

$$(A-2I)(A+I) = \begin{bmatrix} 0 & 0 & 0 \\ a & 0 & 0 \\ b & c & -3 \end{bmatrix} \cdot \begin{bmatrix} 3 & 0 & 0 \\ a & 3 & 0 \\ b & c & 0 \end{bmatrix} = 0.$$

$$\text{Calculating gives } = \begin{bmatrix} 0 & 0 & 0 \\ 3a & 0 & 0 \\ ac & 0 & 0 \end{bmatrix}$$

therefore, $a=0$, $b \in \mathbb{R}$, $c \in \mathbb{R}$ are all possible (a, b, c) .

2025. Fall.

4. Let $N \subseteq GL_n(\mathbb{R}) \times GL_n(\mathbb{R})$ be the set of all pairs (A, B) such that $\det A = \pm \det B$.

20m

Show that N is normal and determine the factor group $(G \times G)/N$.

Sol). Let $(A, B) \in N$ and $(C, D) \in GL_n(\mathbb{R}) \times GL_n(\mathbb{R})$ be given. Then

$$(C, D)(A, B)(C, D)^{-1} = (CAC^{-1}, DBD^{-1})$$

And, since the determinant preserves the multiplication of matrix,

$$\det CAC^{-1} = \det C \cdot \det C^{-1} \cdot \det A = \det A$$

$$\text{and } \det DBD^{-1} = \det D \cdot \det D^{-1} \cdot \det B = \det B$$

So, since $(A, B) \in N$, $\det CAC^{-1} = \det A = \pm \det B = \pm \det DBD^{-1}$

therefore $(C, D)(A, B)(C, D)^{-1} \in N$. And, to show N is subgroup,

first, $(I, I) \in N$ trivially, so $N \neq \emptyset$. Let $(A, B), (C, D) \in N$ be given. Then

$$(A, B) \cdot (C, D)^{-1} = (AC^{-1}, BD^{-1})$$

satisfies $\det AC^{-1} = \det A \cdot \det C^{-1} = \pm \det B \cdot \det C^{-1} = \pm \det B \cdot \det D^{-1} = \pm \det BD^{-1}$.

therefore, by the subgroup criterion, N is subgroup, and normal. $\perp$ hom.

Define a map

$$\mathcal{Q}: GL_n(\mathbb{R}) \times GL_n(\mathbb{R}) \rightarrow \mathbb{R}^{\times}; (A, B) \mapsto (\det A - \det B^{-1})^2$$

Then, $\ker \mathcal{Q} = N$ and $\mathcal{Q}$ is homomorphism, onto $(0, \infty)$, because

the det preserves multiplication, so is $\mathcal{Q}$, and onto is given by: fix $B = I$, and $A = \text{diag}(r, 1, \dots, 1)$ where $r > 0$ is arbitrary.

By the first isomorphism theorem,

$$GL_n(\mathbb{R}) \times GL_n(\mathbb{R}) / N \cong ((0, \infty), \times) \quad \perp \text{ hom.} \quad \textcircled{2}$$

Comment: $\mathcal{Q}$ 은 먼저 $\mathbb{R}^{\times}$ 라고 시작하고, hom 임을 보려면 앞 계산을 꼭 해야 함!

12-25. Fall

10m

5. Let G be an order 21 subgroup of $\mathbb{C}^\times$.

a) G is cyclic.

b). For any $g \in G$, $x^{12} = g^{15}$ has 3 distinct solutions in G .

Proof. a). Since $\mathbb{C}$ is abelian, so G is also abelian.

By the Cauchy's theorem, G has order 3 element x and order 7 element y .
Now, since 3 and 7 are relatively prime, $|xy| = 21$, so $G = \langle xy \rangle$.

b). By a), $G \cong \mathbb{Z}/21\mathbb{Z}$. So, we can write the given equation as

$$12x \equiv 15g \pmod{21} \quad (g \in \mathbb{Z}/21\mathbb{Z}).$$

Regardless of g , $\gcd(12, 21) = 3$ divides $15g$, so it has three distinct solutions.

Note: $ax \equiv b \pmod{n}$ has a solution iff $\gcd(a, n) \mid b$.

And in such a case, it has $\gcd(a, n)$ solutions.

2025. Fall.

6. Let $\{a_n\}$ be a bounded sequence in $\mathbb{R}$. Define

22m

$$\{a_n\}^* = \{x \in \mathbb{R} \mid x \text{ is a subsequential limit of } \{a_n\}\}$$

a). Prove that $\{a_n\}^*$ is closed.

b). Show $\limsup_{n \rightarrow \infty} a_n = \sup \{a_n\}^*$

c). Is there a sequence $\{b_n\}$ such that $\{b_n\}^* = [0, 1]$?

Proof a). If $\{a_n\}^*$ is finite, then clearly closed in $\mathbb{R}$.

Assume that $\{a_n\}^*$ is infinite. Since $\{a_n\}$ is bounded, so the set of subsequential limits $\{a_n\}^*$ is bounded set. So it has a limit point.

Let α be a limit point of $\{a_n\}^*$. Then, given $\epsilon \in \mathbb{R}$, there exists a

subsequential limit p of $\{a_n\}$ such that $p \in B_{\frac{\epsilon}{2}}(\alpha) \setminus \{\alpha\}$.

For $k=1$, choose such a $p_1 \in B_{\frac{\epsilon}{2}}(\alpha) \setminus \{\alpha\}$, and since p is a subsequential limit, choose $a_{p_1} \in B_{\frac{1}{k}}(p)$, then $a_{p_1} \in B_{\frac{\epsilon}{2}}(\alpha)$.

For chosen $a_{p_1}, \dots, a_{p_{k-1}}$, choose $p_k \in B_{\frac{\epsilon}{2}}(\alpha) \setminus \{\alpha\}$.

Then, similarly, there is $a_{p_k} \in B_{\frac{\epsilon}{2}}(\alpha)$ such that $p_k > p_{k-1}$.

because no such p_k exists, then it is contradiction to a_{p_k} is a subsequential limit. Therefore, the arbitrary limit point α is contained in $\{a_n\}^*$,

so it is closed.

b). Since a), $\{a_n\}^*$ is closed and bounded, so compact. Thus

$$\sup \{a_n\}^* = \max \{a_n\}^* = \alpha \text{ for some subsequential limit } \alpha.$$

So, given $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that $n \geq N \Rightarrow a_n < \alpha + \epsilon$ (since $\alpha = \max \{a_n\}^*$)

and there exists infinitely many terms in $\{a_n\}$ such that $\alpha - \epsilon < a_n$ (since $a_{p_k} \rightarrow \alpha$)

Thus $\limsup_{n \rightarrow \infty} a_n = \alpha$

c). Let b_n be defined as:

$$\frac{1}{1}, \frac{1}{2}, \frac{2}{2}, \frac{1}{3}, \frac{2}{3}, \frac{3}{3}, \dots$$

Then, given $b \in [0, 1]$, there exists a subsequence in $\{b_n\}$ converges to b , because: for any $\epsilon > 0$, there are infinitely many terms in b_n such that

$$|b_n - b| < \epsilon,$$

so we can construct the subsequence. (20)

2025, Fall.

7. Let E be a nonempty bounded subset of a Metric space (X, d) .

10m

Define $\text{diam } E := \sup\{d(x, y) \mid x, y \in E\}$

a). Prove $\text{diam } E = \text{diam } \bar{E}$

b). If E is compact, then $d(x, y) = \text{diam } E$ for some $x, y \in E$.

Proof.

a). Since $E \subseteq \bar{E}$, $\text{diam } E \leq \text{diam } \bar{E}$ is clear.

To show the inverse inequality, using the property of supremum.

Given $\varepsilon > 0$, there exists $x, y \in \bar{E}$ such that

$$\text{diam } \bar{E} - \frac{\varepsilon}{2} < d(x, y) \leq \text{diam } \bar{E}$$

And, since $\bar{E}$ is closure, there are $x', y' \in E$ such that

$$d(x', x) < \frac{\varepsilon}{4} \quad \text{and} \quad d(y, y') < \frac{\varepsilon}{4}.$$

Now, $\text{diam } \bar{E} < d(x, y) + \frac{\varepsilon}{2} < d(x, x') + d(x', y') + d(y', y) + \frac{\varepsilon}{2} < d(x', y') + \varepsilon$.

Since $\varepsilon > 0$ was arbitrary, $\text{diam } \bar{E} \leq \text{diam } E$, so proved. 7m.

b). Since E is compact set, so $E \times E$ is also compact set.

Since the metric function d is continuous, it preserves compactness, so $d[E \times E] \subseteq \mathbb{R}$ is compact set. Therefore,

$$\text{diam } E = \max d[E \times E] \in d[E \times E],$$

so there is $(x', y') \in E \times E$ such that $d(x', y') = \max d[E \times E]$ 3m.

2025 Fall.

8. Let $a > 1$. Evaluate

5m

$$\lim_{t \rightarrow 0^+} \int_t^{at} \frac{\cos x}{x} dx$$

Proof. Consider the Taylor expansion

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$$

Therefore $\frac{\cos x}{x} = \frac{1}{x} - \frac{x}{2!} + \frac{x^3}{4!} - \dots$

Now,
$$\int_t^{at} \frac{\cos x}{x} dx = \int_t^{at} \frac{1}{x} + \int_t^{at} -\frac{x}{2!} + \frac{x^3}{4!} - \dots dx$$
$$= \ln a + \sum_{n=1}^{\infty} \int_t^{at} (-1)^n \frac{x^{2n-1}}{(2n)!} dx$$

(It is possible, because $\sum (-1)^n x^{2n-1}/(2n)!$ converges for all $x \in \mathbb{R}$)

Now,
$$\lim_{t \rightarrow 0^+} \sum_{n=1}^{\infty} \int_t^{at} (-1)^n \frac{x^{2n-1}}{(2n)!} dx = 0,$$
 so

the given limit is $\ln a$.

Revised in next

Observation for $\lim_{t \rightarrow 0^+} \int_t^a \frac{\cos x}{x} dx$.

As we know, the power series representation

$$\frac{1}{x} \cdot \cos x = \frac{1}{x} \cdot \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = \frac{1}{x} + \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n-1}}{(2n)!}$$

is valid for all $x \neq 0$. Therefore, for any $t > 0$,

$$\begin{aligned} \int_t^a \frac{\cos x}{x} dx &= \int_t^a \frac{1}{x} + \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n-1}}{(2n)!} dx \\ &= \int_t^a \frac{1}{x} dx + \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} \Big|_t^a \end{aligned}$$

Meanwhile, the series converges uniformly in $[t, a]$ for small enough $t > 0$, specifically, $t < \frac{1}{a}$, then for each $n \in \mathbb{N}$,

$$\left| (-1)^n \frac{x^{2n}}{(2n)!} \right| \leq \left| \frac{1}{(2n)!} \right| \quad \text{and} \quad \sum_{n=1}^{\infty} \frac{1}{(2n)!} < \infty,$$

therefore the Weierstrass M-test gives the condition,

Thus, we can exchange the integral to the sum,

$$\int_t^a \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n-1}}{(2n)!} = \sum_{n=1}^{\infty} \int_t^a (-1)^n \frac{x^{2n-1}}{(2n)!} = \sum_{n=1}^{\infty} (-1)^n \left(\frac{1}{2n \cdot (2n)!} \cdot (a^{2n} - t^{2n}) \right)$$

Converges uniformly similarly, now, we can exchange the limit to the sum

$$\begin{aligned} &\lim_{t \rightarrow 0^+} \sum_{n=1}^{\infty} (-1)^n \frac{1}{2n \cdot (2n)!} \cdot t^{2n} (a^{2n} - 1) \\ &= \sum_{n=1}^{\infty} \lim_{t \rightarrow 0^+} \quad // \\ &= \sum_{n=1}^{\infty} 0 \\ &= 0 \end{aligned}$$

Consequently, the given integral is

$$\int_t^a \frac{1}{x} dx = \ln a - \ln t = \ln a.$$

Note: $\frac{\cos x}{x} = \frac{1}{x} + \frac{\cos x - 1}{x}$, $\left| \int_t^a \frac{\cos x - 1}{x} dx \right| \leq \int_t^a \frac{x}{2} dx \rightarrow 0$ as $t \rightarrow 0$.

2025. Fall.

12. Define an equi. rel on $\mathbb{R}$ as $x \sim y \Leftrightarrow x - y \in \mathbb{Q}$.

" as $x \sim y \Leftrightarrow x = y$ or $x - y \in \mathbb{Q}$.

a). Describe the quotient topology on $\mathbb{R}/\sim$.

b). " on $\mathbb{R}/\sim$.

c). In $\mathbb{R}/\sim$, determine if every point is closed.

d). " the image of the set of irrationals under quotient map is compact.

Proof. Since $\mathbb{Q}$ is closed under the operation $-$, so these relations are equi. rel.

a). First, $\mathbb{R}/\sim = \{a + \mathbb{Q} \mid a \in \mathbb{R}\}$, where $a + \mathbb{Q} = \{a + q \mid q \in \mathbb{Q}\}$. And, given a basis element $(a, b) \in \mathbb{R}$, for the projection $\pi: \mathbb{R} \rightarrow \mathbb{R}/\sim: a \mapsto a + \mathbb{Q}$.

$$\pi^{-1}[\pi[(a, b)]] = \bigcup_{r \in (a, b)} r + \mathbb{Q} = \bigcup_{q \in \mathbb{Q}} (a + q, b + q)$$

is open and since π is quotient map, so $\pi[(a, b)]$ is open, so π is open. Therefore, the topology on $\mathbb{R}/\sim$ is generated by

$$\mathcal{B}_{\sim} = \{\pi[(a, b)] \mid a < b\} = \{(a, b) + \mathbb{Q} \mid a < b\}.$$

But, by the density of rationals, for any $a < b$, $(a, b) + \mathbb{Q} = \mathbb{R}/\mathbb{Q}$. Therefore the topology on $\mathbb{R}/\sim$ is indiscrete.

2025, Fall.

13. Consider a regular plane curve $C: I \rightarrow \mathbb{R}^2$.

a). Give a formula for the normal n to C , and define the signed curvature.

b). Explain the relation between the total angle change and the signed curvature.

c). Assume C is a closed plane curve parametrized by arc-length, and $I = [0, 2\pi]$.

Prove or disprove that: there are $t, t' \in I$ such that $C'(t) = -C'(t')$.

and if $t=0$, there is such a $t' \in I$?

Moreover, show that $\int_0^{2\pi} |K(s)| ds \geq 2\pi$ where $K(s)$ is the signed curvature.

Proof. Denote $C(t) = (x(t), y(t))$.

a). The tangent vector is

$$T(t) = \frac{C'(t)}{|C'(t)|} = \frac{(x'(t), y'(t))}{\sqrt{x'(t)^2 + y'(t)^2}}$$

So, the normal vector is

$$N(t) = \frac{(-y'(t), x'(t))}{\sqrt{x'(t)^2 + y'(t)^2}}$$

And, since the curvature $K(s(t))$ satisfies $T'(t) = K(t) \cdot N(t)$,

$$K(t) = K(t) \langle N(t), N(t) \rangle = \langle T'(t), N(t) \rangle$$

Pebris,

Denote $C = (x, y)$. Then,

$$T = (x', y') / |(x', y')| = \frac{(x', y')}{\sqrt{x'^2 + y'^2}}$$

is the tangent vector, and the normal vector is

$$N = \frac{(-y', x')}{\sqrt{x'^2 + y'^2}}$$

Moreover, the signed curvature K satisfies $K N = T'$,

so $K = K \cdot \langle N, N \rangle = \langle T', N \rangle \cdot |C'(t)|^{-1}$ Meanwhile

$$T' = \frac{(x'', y'')}{\sqrt{x'^2 + y'^2}} + (x', y') \cdot \left(-\frac{1}{2} \cdot \frac{2x'x'' + 2y'y''}{(x'^2 + y'^2)^{\frac{3}{2}}} \right) \text{ so}$$

$$K = \frac{1}{x'^2 + y'^2} \cdot (x'y'' - x''y') \cdot \frac{1}{\sqrt{x'^2 + y'^2}}$$

TSNU. Q. 161.

Prove that $\lim_{b \rightarrow \infty} \int_0^b \frac{\sin x}{x} dx$ exists.

Proof. First, let $a_n = n\pi$ ($n \in \mathbb{N}$). Then, a sequence $\left\{ \int_0^{a_n} \frac{\sin x}{x} dx \right\}$ converges, because:

$\left| \int_{n\pi}^{(n+1)\pi} \frac{\sin x}{x} dx \right|$ is decreasing and converges to zero, and

$\int_{n\pi}^{(n+1)\pi} \frac{\sin x}{x} dx$ is positive if n is even and is negative if n is odd.

That is, $\int_0^{n\pi} \frac{\sin x}{x} dx = \sum_{k=1}^n \int_{a_{k-1}}^{a_k} \frac{\sin x}{x} dx$ is alternating series, so converges.

7226. Prima.

3. Let V be an n -dimensional vector space over $\mathbb{C}$,
Let $T: V \rightarrow V$ be a linear operator such that $T^3 = T$.

27m

a). Find all possible characteristic polynomials and corresponding minimal polynomials for T .

b). Prove $V = \ker T \oplus \operatorname{Im} T$.

Proof.

a). Since $T^3 - T = 0$, the minimal polynomial $m(t)$ for T divides $t^3 - t = 0$.

$t^3 - t = t(t-1)(t+1)$, there are possibilities:

$$m(t) = t \text{ or } t-1 \text{ or } t+1 \text{ or } t(t-1) \text{ or } t(t+1) \text{ or } (t-1)(t+1) \text{ or } t(t-1)(t+1).$$

Since $m(t)$ divides the characteristic polynomial and they have same characteristic values, the possible characteristic polynomial is:

$$t^{r_1} \cdot (t-1)^{r_2} \cdot (t+1)^{r_3} \text{ where } r_1, r_2, r_3 \text{ are non-negative integers and } r_1 + r_2 + r_3 = n.$$

If $r_1 = 0$ and $r_2, r_3 \neq 0$, then the corresponding minimal polynomial is $(t-1)(t+1)$

and other cases are similar.

b). ~~Given $\alpha \in \ker T \cap \operatorname{Im} T$, $T\alpha = 0$ and $\alpha = T\beta$ for some $\beta \in V$.~~

Given $\alpha \in V$, $\alpha = T^2\alpha + \alpha - T^2\alpha$ and since $T(\alpha - T^2\alpha) = (T - T^3)\alpha = 0\alpha = 0$,
so $\alpha - T^2\alpha \in \ker T$ and trivially $T^2\alpha \in \operatorname{Im} T$. Moreover, if $T\alpha = 0$ and $\alpha \in \operatorname{Im} T$,
Then $\alpha = T\beta = T^3\beta$ for some $\beta \in V$ and $T\alpha = T^2\beta = 0$ so $T^3\beta = TT^2\beta = \alpha = 0$,
so $V = \ker T \oplus \operatorname{Im} T$.

Modified.

Let T be a linear operator on the fin. dim space V over $\mathbb{C}$ satisfying $T^4 = T^2$.

a). Prove that $V = \ker T^2 \oplus \operatorname{Im} T^2$.

b). Describe $T|_{\operatorname{Im} T^2}$.

c). Prove that T is diagonalizable $\Leftrightarrow \ker T = \ker T^2$.

observe: $T^4 - T^2 = 0$, so the minimal polynomial divides $t^2(t-1)(t+1)$.

a). Given $\alpha \in V$, $\alpha = T^2\alpha + \alpha - T^2\alpha$ and $T^2\alpha \in \operatorname{Im} T^2$ clearly, and $T^2(I - T^2)\alpha = (T^2 - T^4)\alpha = 0 \cdot \alpha = 0$, so $\alpha - T^2\alpha$, which proves $\alpha - T^2\alpha \in \ker T^2$, so $V = \ker T^2 + \operatorname{Im} T^2$. Moreover, given $\alpha \in \ker T^2 \cap \operatorname{Im} T^2$, $T^2\alpha = 0$ and $\alpha = T^2\beta$ for some $\beta \in V$, so

$$\alpha = T^2\beta = T^4\beta = T^2T^2\beta = T^2\alpha = 0$$

so $V = \ker T^2 \oplus \operatorname{Im} T^2$.

b). Given $\alpha \in \operatorname{Im} T^2$, $\alpha = T^2\beta$ for some $\beta \in V$. so, $T\alpha = T^3\beta$.

Multiplying T , then $T^2\alpha = T^4\beta = T^2\beta$, so $T^2(\alpha - \beta) = 0$.

Therefore, $\alpha - \beta \in \ker T^2$, but since $\alpha \in \operatorname{Im} T^2$ and $\operatorname{Im} T^2 \cap \ker T^2 = 0$, $\beta = \alpha + \gamma$, where $\gamma \in \ker T^2$. Now, $T\alpha = T^3\beta = T^3\alpha + T^3\gamma = T^3\alpha$, so, $T|_{\operatorname{Im} T^2} = T^3|_{\operatorname{Im} T^2}$, moreover, since $T^2 = T^4$, $T^2|_{\operatorname{Im} T^2} = I$.

c). Since T is diagonalizable $\Leftrightarrow$ minimal polynomial of T splits and separable. That is, the minimal polynomial is $t^a(t-1)^b(t+1)^c$, $a, b, c \in \{0, 1\}$.

2026 Spring

5. Let $f(x) = x^4 + 2x + 3 \in \mathbb{F}_7[x]$ and $K = \mathbb{F}_7[x]/(f)$.

Let $\mathcal{Q}: K \rightarrow K: a \mapsto a^7$.

a). Prove K is a field and $\mathcal{Q}$ is Automorphism.

b). Let $G = \langle \mathcal{Q} \rangle \leq \text{Aut}(K)$. Determine the number of $a \in K$ such that $|\text{Orb}(a)| = 2$.

Proof

a). Since $f(0)=3, f(1)=6, f(2)=2, f(3)=6, f(4)=1, f(5)=1, f(6)=2$, so $f(x)$ has no linear factor. So, if $f(x)$ is reducible,

$$f(x) = (x^2 + ax + b)(x^2 + cx + d) \text{ for some } a, b, c, d \in \mathbb{F}_7.$$

Then,
$$\begin{cases} a+c=0 & \text{If } a=0 \text{ or } c=0, \text{ then } a=c=0, \text{ so } 0=2. \\ ac+b+d=0 & \text{If } b=0 \text{ or } d=0, \text{ then } 0=3, \\ ad+bc=2 & \text{Therefore, } a, b, c, d \text{ are non-zero.} \\ bd=3 & \text{Reduce the system:} \end{cases}$$

$$\Rightarrow \begin{cases} b+d=a^2 \\ a(d-b)=2 \\ bd=3 \end{cases}$$

If $b=d$, then $b^2=3$, but $\sqrt{3} \notin \mathbb{F}_7$, so $b \neq d$.

And, $3 = 1 \cdot 3$ or $2 \cdot 5$ or $4 \cdot 6$, so

$a^2 = 4$ or 0 or 3 . Since $\sqrt{3} \notin \mathbb{F}_7$ and $a \neq 0$, $a^2 = 4$, so $a = 2$ or 5 ,

with $\{b, d\} = \{1, 3\}$. But, $2 \cdot (1-3) = 3 \neq 2$,
 $2 \cdot (3-1) = 4 \neq 2$,
 $5 \cdot (1-3) = 4 \neq 2$,
 $5 \cdot (3-1) = 3 \neq 2$

So contradiction. Therefore, f must be an irreducible.

Since $\mathbb{F}_7$ is a field, $\mathbb{F}_7[x]$ is p.z.d., so (f) is maximal, so $\mathbb{F}_7[x]/(f)$ is a field. And, since the characteristic of K is 7, for any $a, b \in K$,

$$(a+b)^7 = \sum_{k=0}^7 \binom{7}{k} a^{7-k} b^k = a^7 + b^7$$

and $(ab)^7 = a^7 b^7$, so $\mathcal{Q}$ is Homomorphism. Moreover,

$\mathcal{Q}(a) = 0 \Leftrightarrow a^7 = 0 \Leftrightarrow a = 0$, so $\ker \mathcal{Q}$ is trivial, so $\mathcal{Q}$ is 1-1.

Now, since $[K: \mathbb{F}_7] = \deg f = 4$, so $\#K = 7^4$, finite, so $\mathcal{Q}$ is bijective.

b). Since every finite field is perfect, so K is Galois over $\mathbb{F}_7$,
so $|\text{Aut}(K)| = \deg f = 4$,

Consider an Action

$$G \times K \rightarrow K : (\mathbb{Q}^n, \alpha) \mapsto \mathbb{Q}^n(\alpha).$$

So $|\text{Orb}(\alpha)| = 2 \iff \mathbb{Q}^2(\alpha) = \mathbb{Q}$ and $\alpha \neq 0 \iff \alpha^{4q} = \alpha$ and $\alpha \neq 0$
So, the fixed field by $\langle \mathbb{Q}^2 \rangle$, $\langle \mathbb{Q}^2 \rangle \setminus \{0\}$ is we desired.

Revised in the next

Revised.

Prove that $f(x) = x^4 + 2x + 3$ is irreducible over $\mathbb{F}_7$.

And describe the Galois group of $f(x)$.

1). $f(x)$ has no linear factor.

$f(0) = 3 \neq 0$. And, since $\mathbb{F}_7^\times$ is a group of order 6 under the multiplication. Therefore, for any $\alpha \in \mathbb{F}_7^\times$, $\alpha^6 = (\alpha^3)^2 = 1$ implies $\alpha^3 = \pm 1$.

If $\alpha^3 = 1$, then $\alpha^4 = \alpha$, so $f(\alpha) = \alpha^4 + 2\alpha + 3 = 3(\alpha + 1) = 0$ implies $\alpha = -1$ but $(-1)^3 = -1 = 6 \neq 1$, so $\alpha^3 \neq 1$.

If $\alpha^3 = -1$, then $\alpha^4 = -\alpha$, so $f(\alpha) = \alpha + 3 = 0$ implies $\alpha = 4$, but $4^3 = 64 = 1 \neq -1$, so $\alpha^3 \neq -1$. Now, $f(x)$ has no root in $\mathbb{F}_7^\times$.

2). $f(x)$ has no factor of degree 2.

Suppose that $f(x) = (x^2 + ax + b)(x^2 + cx + d)$. Then, we have a system:

$$\begin{cases} a+c=0 \\ ac+b+d=0 \\ ad+bc=2 \\ bd=3 \end{cases} \Rightarrow \begin{cases} b+d=a^2 \\ a(d-b)=2 \\ bd=3 \end{cases} \Rightarrow \begin{cases} (d-b)^2 = (b+d)^2 - 4bd = a^4 - 12 = a^4 + 2 \\ a^2(d-b)^2 = 4 \end{cases}$$

$\alpha \neq 0$, so $\alpha^6 = 1$

$\Rightarrow \alpha^6 + 2\alpha^2 = 1 + 2\alpha^2 = 4 \Rightarrow \alpha^2 = 5$. But, $\{x^2 \mid x \in \mathbb{F}_7\} = \{0, 1, 4, 2\}$, so $\sqrt{\alpha} \notin \mathbb{F}_7$.

Now, f is irreducible $\Rightarrow \mathbb{F}_7[x]/(f)$ is a field.

2026 Fall.

4. Let G be a group and $H_1, H_2 \leq G$ be subgroups of finite index.

a). Prove that the map

$$T: G \rightarrow S_{G/H_1 \times G/H_2} : x \mapsto T_x \text{ where } T_x(aH_1, bH_2) = (xaH_1, xbH_2)$$

is Homomorphism and $\text{Ker } T \subseteq H_1 \cap H_2$.

b). Prove $H_1 \cap H_2$ has finite index.

Proof

$$\begin{aligned} \text{a). Given } g_1, g_2 \in G, T(g_1 g_2)(aH_1, bH_2) &= (g_1 g_2 aH_1, g_1 g_2 bH_2) \\ &= T(g_1) \circ T(g_2)(aH_1, bH_2) \end{aligned}$$

for each $a, b \in G$, so it preserves the operation.

$$\text{And } g \in \text{Ker } T \Leftrightarrow T_g = \text{id} \Leftrightarrow \text{for all } a, b \in G, T_g(aH_1, bH_2) = (gaH_1, gbH_2) = (aH_1, bH_2)$$

$$\begin{aligned} &\Leftrightarrow \text{for all } a, b \in G, a^{-1}ga \in H_1 \text{ and } b^{-1}gb \in H_2 \\ &\Rightarrow g \in H_1 \cap H_2 \text{ (take } a=b=1_G) \end{aligned}$$

b). By the first-isomorphism theorem,

$$G/\text{Ker } T \cong \text{Im } T \leq S_{G/H_1 \times G/H_2}$$

and since H_1, H_2 have finite index, so the Codomain of T is finite set.

$$\text{Now, } |G : H_1 \cap H_2| \leq |G : \text{Ker } T| \leq \# S_{G/H_1 \times G/H_2}$$

because $\text{Ker } T \subseteq H_1 \cap H_2$, so proved.

Note: If $A \leq B \leq G$ and $|G:A|$ is finite, then

$$G/A \rightarrow G/B : gA \mapsto gB$$

is well-defined and onto, because: Since $A \leq B$,

$$g_1 A = g_2 A \Leftrightarrow g_1 g_2^{-1} \in A \leq B \Rightarrow g_1 g_2^{-1} \in B \Leftrightarrow g_1 B = g_2 B$$

and surjectiveness is clear.

12.26. Fall.

8. For $f \in C([0,1])$, define its extension

$$\tilde{f}(x) := \begin{cases} f(0) & \text{if } x < 0 \\ f(x) & \text{if } 0 \leq x \leq 1 \\ f(1) & \text{if } x > 1 \end{cases}$$

For each $n \geq 1$, define

$$p_n(x) := \frac{n}{2} \cdot \chi_{[-\frac{1}{n}, \frac{1}{n}]}(x)$$

Define $T_n f(x) := \int_{\mathbb{R}} p_n(x-y) \tilde{f}(y) dy$, $x \in \mathbb{R}$.

a). Show that $T_n f \rightarrow f$ on $[0,1]$.

b). Determine $T_n f \rightarrow f$ uniformly on $[0,1]$ or not.

Proof. Given $x \in \mathbb{R}$ and $n \in \mathbb{N}$, $x-y \in [-\frac{1}{n}, \frac{1}{n}] \Leftrightarrow -\frac{1}{n}+x \leq y \leq \frac{1}{n}+x$. Therefore,

$$T_n f(x) = \int_{-\frac{1}{n}+x}^{\frac{1}{n}+x} \frac{n}{2} \cdot \tilde{f}(y) dy.$$

Since $\tilde{f}$ is continuous on $[-1,2]$, so it is continuously uniformly.

Given $\varepsilon > 0$, take $\delta > 0$ such that $\forall x, t \in [0,1]$, $|x-t| < \delta \Rightarrow |f(x) - f(t)| < \varepsilon$.

Now, given $x \in [0,1]$, and $n > \frac{1}{\delta}$,

$$\begin{aligned} |T_n f(x) - f(x)| &= \left| \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} \frac{n}{2} \tilde{f}(y) dy - \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} \frac{n}{2} f(x) dy \right| \\ &= \left| \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} \frac{n}{2} [\tilde{f}(y) - f(x)] dy \right| \\ &\leq \frac{n}{2} \cdot \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} |\tilde{f}(y) - f(x)| dy \\ &< \frac{n}{2} \cdot \frac{2}{n} \cdot \varepsilon = \varepsilon \end{aligned}$$

Therefore, $\sup_{x \in [0,1]} |T_n f - f| \rightarrow 0$ as $n \rightarrow \infty$, so converges uniformly.

